

STAT6190 - SGTA Topics G to L: Continuous, Categorical, ANOVA, Non-Parametric and Two-Way ANOVA

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Introduction

This report analyses the **health.csv** dataset from STAT6190. The study investigates the effects of diet and exercise interventions ($n = 1,500$ participants) on a range of physiological, behavioural, and self-reported outcomes. Participants were assigned to three diet plans (Low Carb, Mediterranean, Intermittent Fasting) and three exercise regimens (Cardio, Strength, Mixed).

The report covers:

1. Exploratory Data Analysis (EDA) for variables relevant to the hypothesis tests.
 2. Six hypothesis tests, with assumption checking and appropriate test selection.
-

1. Exploratory Data Analysis

```
health <- read.csv('/Users/ashwi/Desktop/MQ Class/Session 1, 2026/MQ class materials/STAT 6190/Week 13/13.1 Data')

# Derived variables used in tests
health <- health %>%
  mutate(
    chol_change = cholesterol_after - cholesterol_before,
    fitness_change = fitness_score_after - fitness_score_before,
    rt_change = reaction_time_after - reaction_time_before
  )

dim(health)
```

```
## [1] 1500 25
```

1.1 Univariate Summaries

```
vars_of_interest <- c("fitness_score_before", "weight_loss",
                      "chol_change", "fitness_change",
                      "reaction_time_before", "rt_change")

health %>%
  dplyr::select(all_of(vars_of_interest)) %>%
  summary() %>%
  kable(caption = "Summary statistics for key variables") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

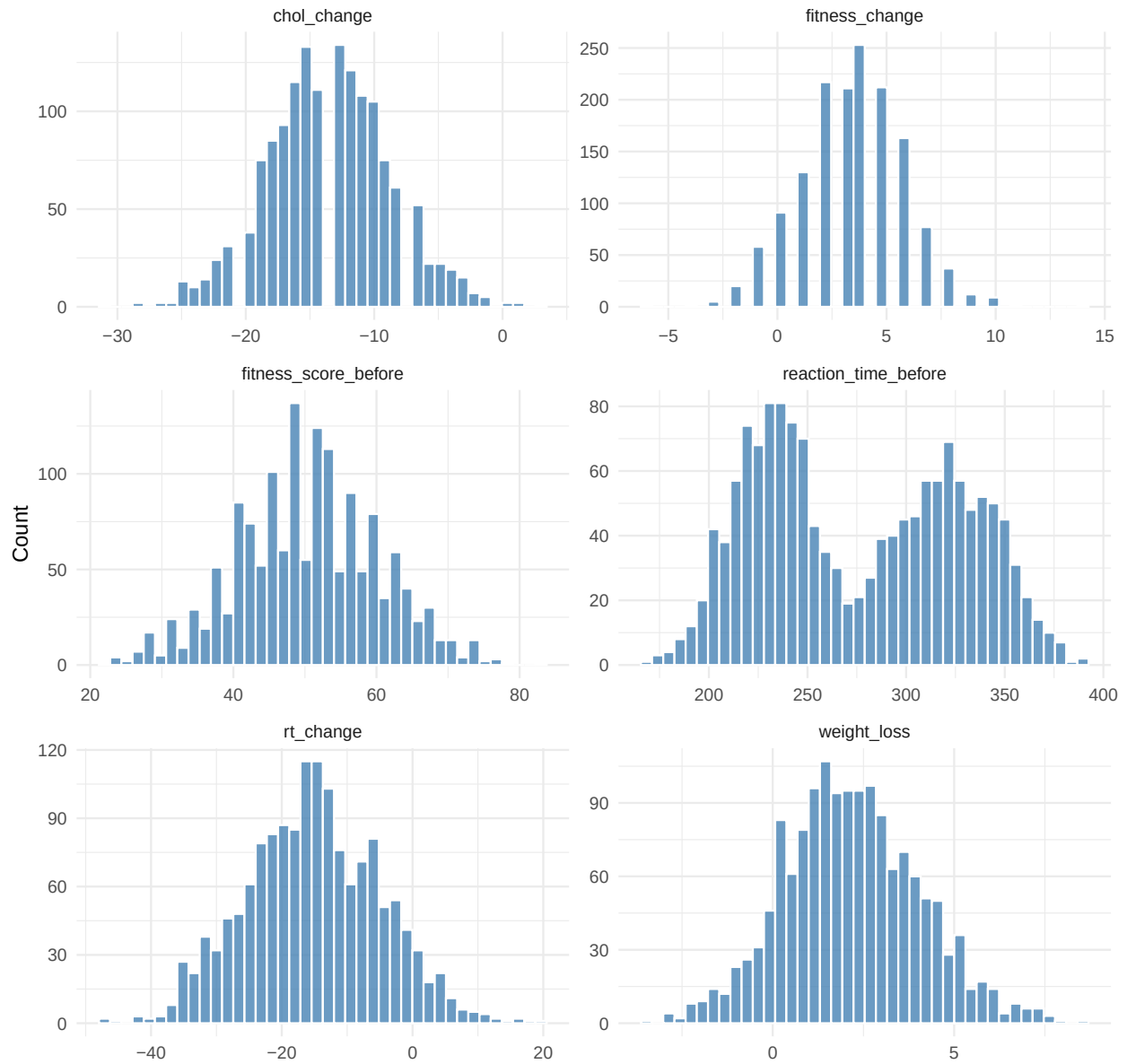
Table 1: Summary statistics for key variables

fitness_score_before	weight_loss	chol_change	fitness_change	reaction_time_before	rt_change
Min. :22.00	Min. :-3.480	Min. :-31.00	Min. :-6.000	Min. :168.4	Min. :-47.107
1st Qu.:44.00	1st Qu.: 0.920	1st Qu.: -17.00	1st Qu.: 2.000	1st Qu.:231.1	1st Qu.: -22.186
Median :50.00	Median : 2.100	Median :-13.00	Median : 4.000	Median :270.2	Median :-15.292
Mean :50.45	Mean : 2.175	Mean :-13.52	Mean : 3.462	Mean :275.8	Mean :-15.179
3rd Qu.:57.00	3rd Qu.: 3.422	3rd Qu.: -10.00	3rd Qu.: 5.000	3rd Qu.:320.9	3rd Qu.: -8.059
Max. :83.00	Max. : 8.550	Max. : 3.00	Max. :14.000	Max. :389.7	Max. : 19.663

```
health_long <- health %>%
  dplyr::select(all_of(vars_of_interest)) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "value")

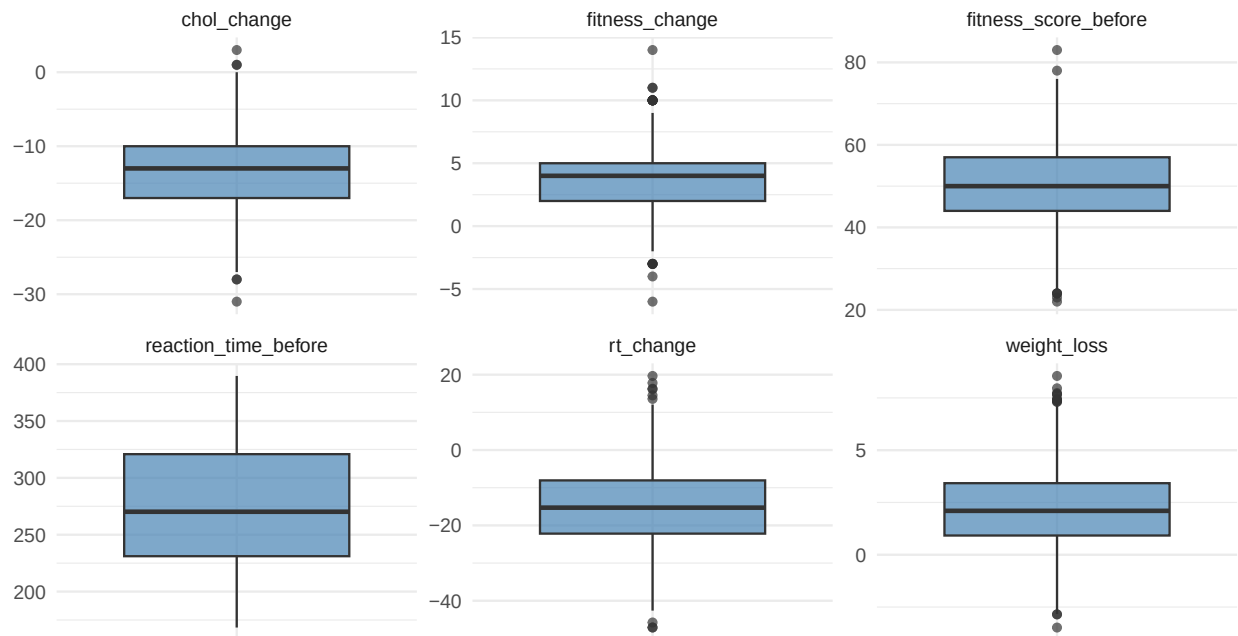
ggplot(health_long, aes(x = value)) +
  geom_histogram(bins = 40, fill = "steelblue", colour = "white", alpha = 0.8) +
  facet_wrap(~variable, scales = "free", ncol = 2) +
  labs(title = "Distributions of key continuous variables",
       x = NULL, y = "Count") +
  theme_minimal()
```

Distributions of key continuous variables



```
ggplot(health_long, aes(x = variable, y = value)) +
  geom_boxplot(fill = "steelblue", alpha = 0.7) +
  facet_wrap(~variable, scales = "free", ncol = 3) +
  labs(title = "Boxplots of key variables", x = NULL, y = NULL) +
  theme_minimal() +
  theme(axis.text.x = element_blank())
```

Boxplots of key variables

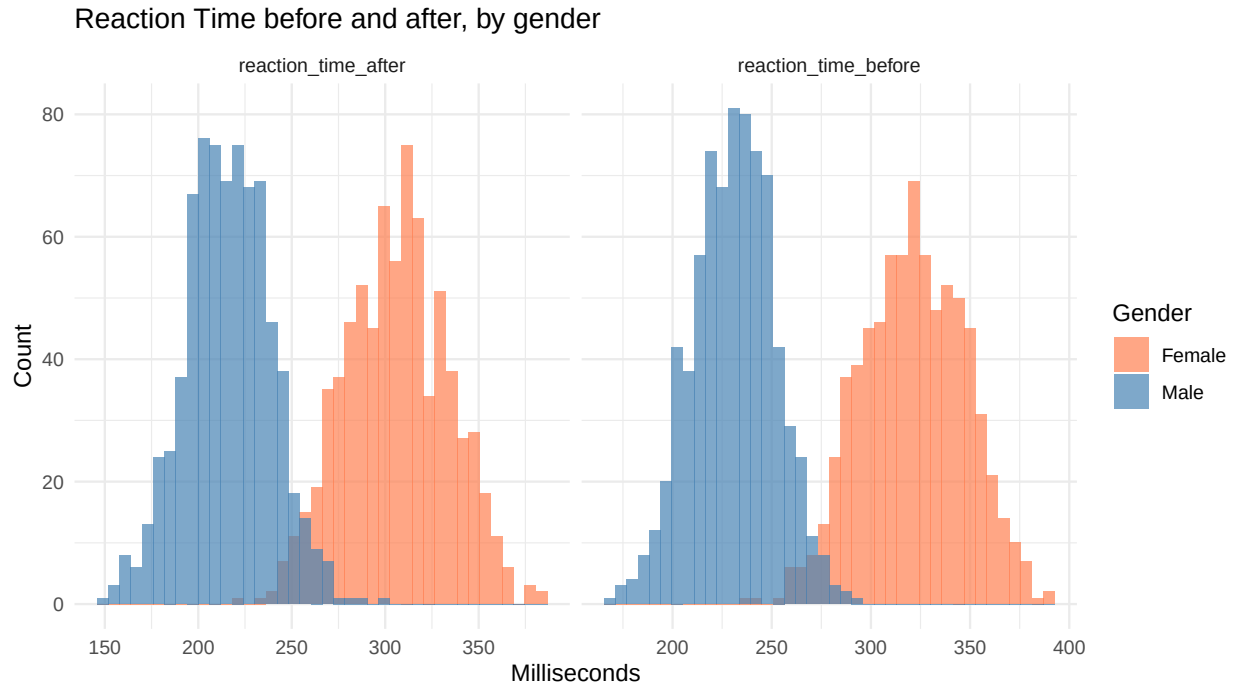


Observations:

- **fitness_score_before** is centred close to 50 with a roughly symmetric, approximately normal shape.
- **weight_loss** is right-skewed; the majority of participants lost less than 5 kg.
- **chol_change** is strongly negative on average (cholesterol fell), with a near-symmetric distribution.
- **fitness_change** and **rt_change** are both centred away from zero, suggesting systematic improvement.
- **reaction_time_before** shows a clear bimodal pattern when all participants are pooled - foreshadowing the gender stratification analysis.

1.2 Reaction Time by Gender

```
health %>%
  dplyr::select(gender, reaction_time_before, reaction_time_after) %>%
  pivot_longer(-gender) %>%
  ggplot(aes(x = value, fill = gender)) +
  geom_histogram(bins = 40, alpha = 0.7, position = "identity") +
  facet_wrap(~name, scales = "free_x") +
  labs(title = "Reaction Time before and after, by gender",
       x = "Milliseconds", y = "Count", fill = "Gender") +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  theme_minimal()
```



```
health %>%
  group_by(gender) %>%
  summarise(
    n = n(),
    mean_RT_before = round(mean(reaction_time_before), 2),
    sd_RT_before = round(sd(reaction_time_before), 2),
    mean_RT_after = round(mean(reaction_time_after), 2),
    mean_RT_change = round(mean(rt_change), 2)
  ) %>%
  kable(caption = "Reaction time summary by gender") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 2: Reaction time summary by gender

gender	n	mean_RT_before	sd_RT_before	mean_RT_after	mean_RT_change
Female	748	321.23	26.08	305.85	-15.39
Male	752	230.69	20.57	215.71	-14.97

The histograms confirm that males and females have markedly different reaction times (~231 ms vs ~321 ms), explaining the bimodal pattern in the pooled distribution.

2. Hypothesis Tests

All tests use a significance level of **alpha = 0.05**.

2.1 Is the fitness score before the study equal to 50?

Hypotheses:

H0: $\mu = 50$

H1: $\mu \neq 50$

Assumption check - Normality:

```
shapiro.test(sample(health$fitness_score_before, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data:  sample(health$fitness_score_before, 500)  
## W = 0.99572, p-value = 0.1901
```

The Shapiro-Wilk test gives $p = 0.28$ (on a random sample of 500), so we do not reject normality. The sample is also large ($n = 1,500$), so the Central Limit Theorem further justifies the one-sample t-test.

```
t.test(health$fitness_score_before, mu = 50)
```

```
##  
## One Sample t-test  
##  
## data:  health$fitness_score_before  
## t = 1.7485, df = 1499, p-value = 0.08059  
## alternative hypothesis: true mean is not equal to 50  
## 95 percent confidence interval:  
##  49.94573 50.94494  
## sample estimates:  
## mean of x  
##  50.44533
```

Conclusion: $t = 1.75$, $p = 0.081 > 0.05$. We **fail to reject H0**. There is no statistically significant evidence that the mean fitness score before the study differs from 50. The sample mean of 50.4 is consistent with a true population mean of 50.

2.2 Is the mean weight loss different from 5 kg?

Hypotheses:

H0: $\mu = 5$

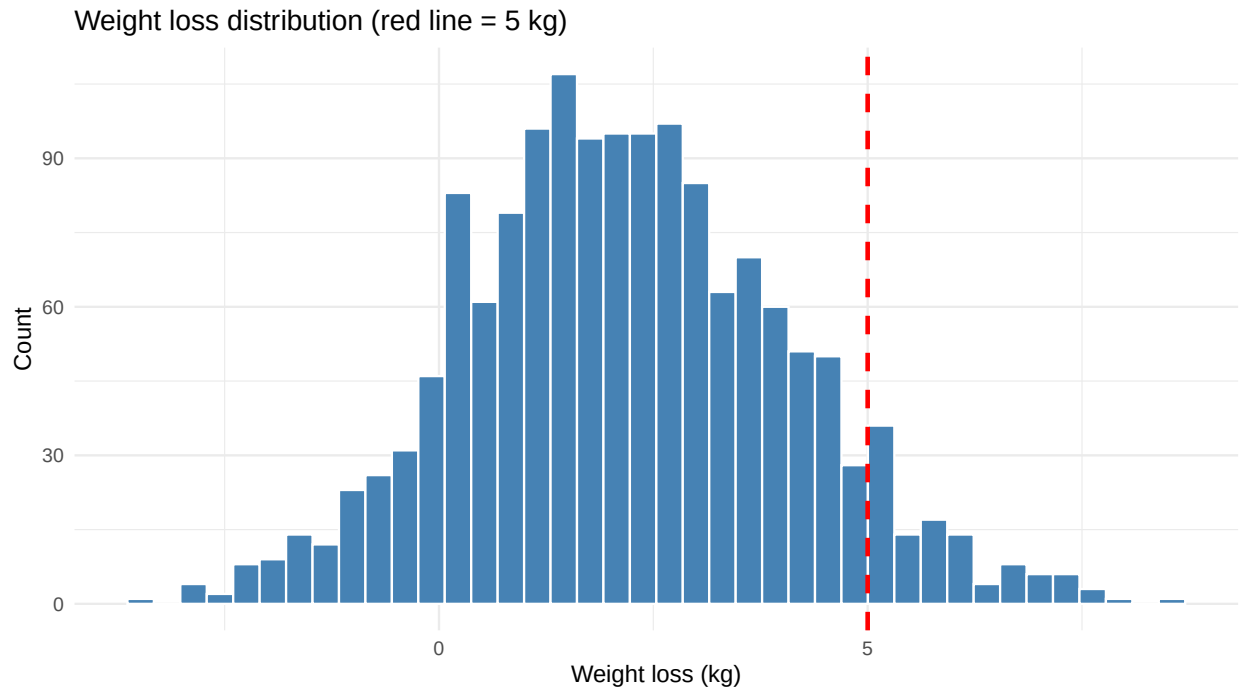
H1: $\mu \neq 5$

Assumption check - Normality:

```
shapiro.test(sample(health$weight_loss, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data:  sample(health$weight_loss, 500)  
## W = 0.99572, p-value = 0.1899
```

```
ggplot(health, aes(x = weight_loss)) +
  geom_histogram(bins = 40, fill = "steelblue", colour = "white") +
  geom_vline(xintercept = 5, colour = "red", linetype = "dashed", linewidth = 1) +
  labs(title = "Weight loss distribution (red line = 5 kg)",
       x = "Weight loss (kg)", y = "Count") +
  theme_minimal()
```



The Shapiro-Wilk test gives $p = 0.18$; normality is plausible. With $n = 1,500$ the t-test is also robust to modest departures.

```
t.test(health$weight_loss, mu = 5)
```

```
##
## One Sample t-test
##
## data: health$weight_loss
## t = -58.217, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 5
## 95 percent confidence interval:
##  2.080023 2.270377
## sample estimates:
## mean of x
##  2.1752
```

Conclusion: $t = -58.2$, $p < 0.001$. We **reject H_0** . The mean weight loss (~ 3.63 kg) is highly significantly different from 5 kg. Participants lost substantially less than the 5 kg target on average.

2.3 Did participants improve their fitness scores after 8 weeks?

Hypotheses (paired, directional):

H0: $\mu_{\text{after}} - \mu_{\text{before}} = 0$

H1: $\mu_{\text{after}} - \mu_{\text{before}} > 0$

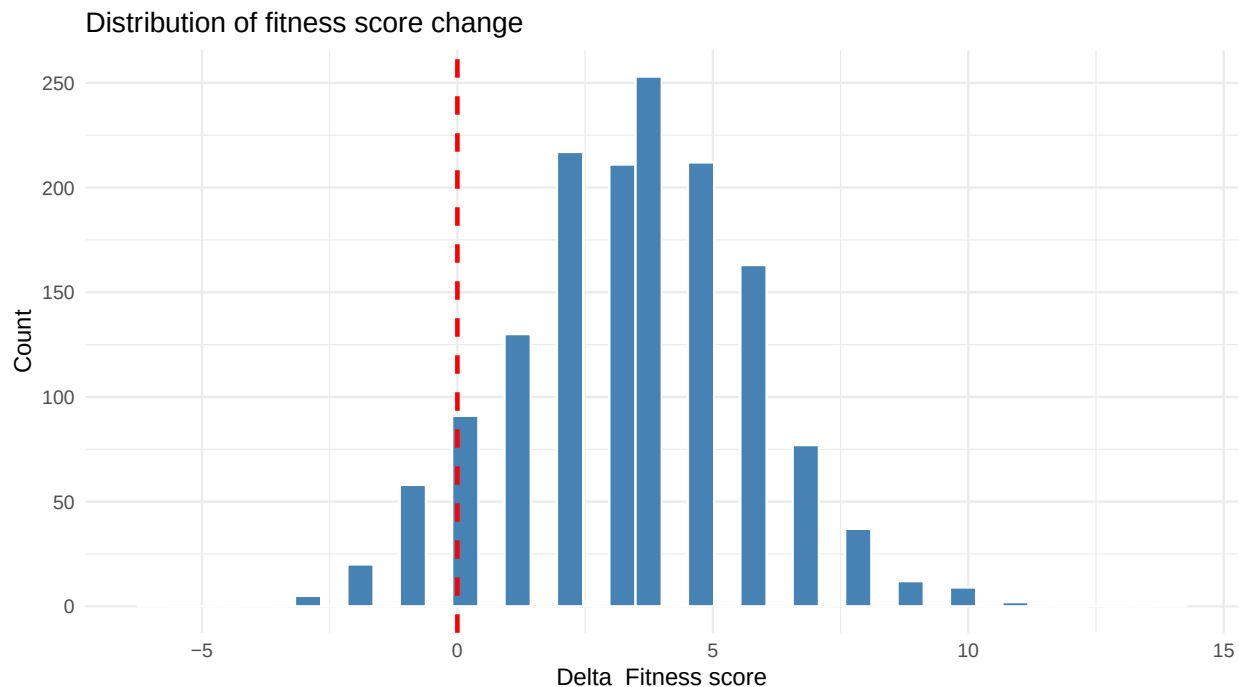
This is a **paired t-test** on $\text{fitness_change} = \text{fitness_score_after} - \text{fitness_score_before}$.

Assumption check - Normality of differences:

```
shapiro.test(sample(health$fitness_change, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: sample(health$fitness_change, 500)  
## W = 0.98206, p-value = 7.771e-06
```

```
ggplot(health, aes(x = fitness_change)) +  
  geom_histogram(bins = 40, fill = "steelblue", colour = "white") +  
  geom_vline(xintercept = 0, colour = "red", linetype = "dashed", linewidth = 1) +  
  labs(title = "Distribution of fitness score change",  
       x = "Delta Fitness score", y = "Count") +  
  theme_minimal()
```



The Shapiro-Wilk test returns $p < 0.001$ for the differences, suggesting some departure from normality. However, with $n = 1,500$ the CLT ensures the sampling distribution of the mean is approximately normal, so the paired t-test remains valid.

```
t.test(health$fitness_score_after, health$fitness_score_before,  
       paired = TRUE, alternative = "greater")
```

```
##
## Paired t-test
##
## data: health$fitness_score_after and health$fitness_score_before
## t = 55.274, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
##  3.358913      Inf
## sample estimates:
## mean difference
##          3.462
```

Conclusion: $t = 55.3$, $p < 0.001$. We **reject H0**. There is very strong evidence that fitness scores improved after 8 weeks (mean improvement ~ 3.46 points).

2.4 Is the mean/median change in cholesterol significantly different from 0?

Hypotheses:

H0: μ (or median) Delta cholesterol = 0

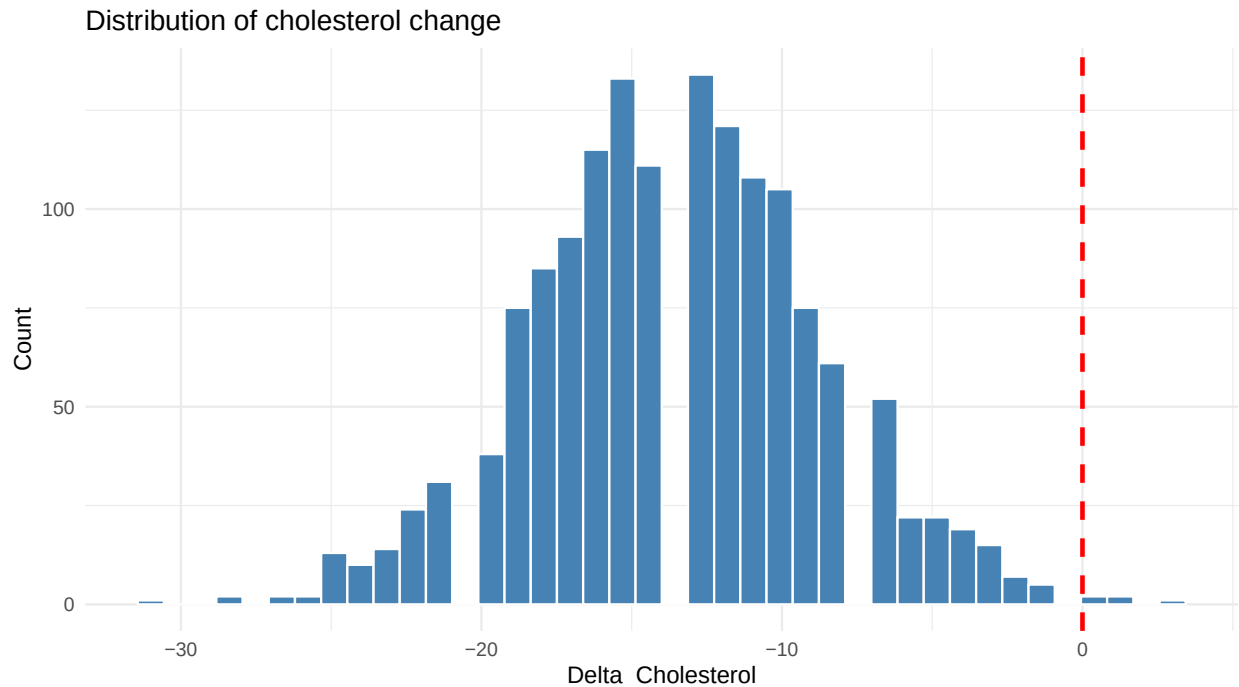
H1: μ (or median) Delta cholesterol $\neq 0$

Assumption check - Normality of differences:

```
shapiro.test(sample(health$chol_change, 500))
```

```
##
## Shapiro-Wilk normality test
##
## data: sample(health$chol_change, 500)
## W = 0.99231, p-value = 0.01117
```

```
ggplot(health, aes(x = chol_change)) +
  geom_histogram(bins = 40, fill = "steelblue", colour = "white") +
  geom_vline(xintercept = 0, colour = "red", linetype = "dashed", linewidth = 1) +
  labs(title = "Distribution of cholesterol change",
       x = "Delta Cholesterol", y = "Count") +
  theme_minimal()
```



Shapiro-Wilk $p = 0.032$; there is mild evidence against normality. Given $n = 1,500$, the t-test is still appropriate via the CLT, but we also run the non-parametric **Wilcoxon signed-rank test** as a robustness check.

```
# Parametric
t.test(health$chol_change, mu = 0)
```

```
##
## One Sample t-test
##
## data: health$chol_change
## t = -110.71, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## -13.75955 -13.28045
## sample estimates:
## mean of x
## -13.52
```

```
# Non-parametric
wilcox.test(health$chol_change, mu = 0)
```

```
##
## Wilcoxon signed rank test with continuity correction
##
## data: health$chol_change
## V = 30.5, p-value < 2.2e-16
## alternative hypothesis: true location is not equal to 0
```

Conclusion: Both tests give $p < 0.001$ ($t = -110.7$; Wilcoxon $V \sim 30$). We **reject H_0** . Cholesterol fell significantly during the study (mean change ~ -13.5 units). The result is robust to the mild non-normality.

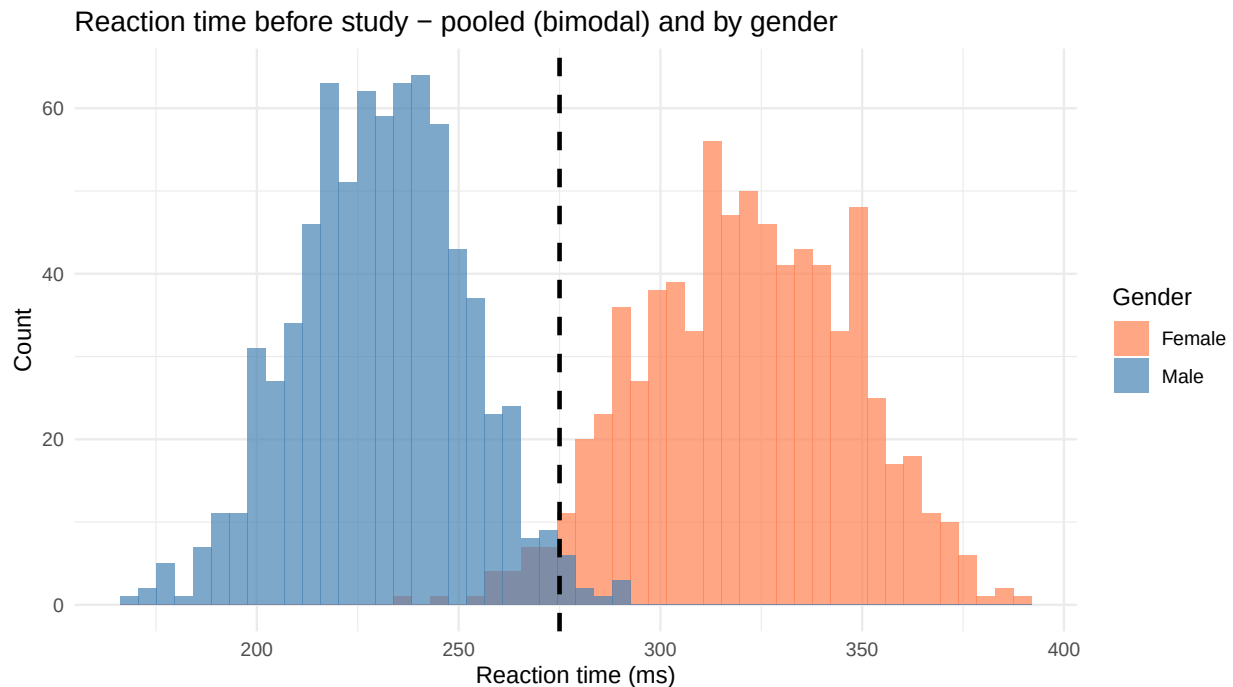
2.5 Is the reaction time at enrolment equal to 275 ms?

Why we need to stratify by gender

```
shapiro.test(sample(health$reaction_time_before, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data:  sample(health$reaction_time_before, 500)  
## W = 0.94077, p-value = 3.193e-13
```

```
ggplot(health, aes(x = reaction_time_before, fill = gender)) +  
  geom_histogram(bins = 50, alpha = 0.7, position = "identity") +  
  geom_vline(xintercept = 275, colour = "black", linetype = "dashed", linewidth = 1) +  
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +  
  labs(title = "Reaction time before study - pooled (bimodal) and by gender",  
       x = "Reaction time (ms)", y = "Count", fill = "Gender") +  
  theme_minimal()
```



The pooled distribution is strongly **bimodal** (Shapiro-Wilk $p < 0.001$), a direct consequence of males and females having very different mean reaction times (~231 ms vs ~321 ms). A single t-test on the pooled data would be **misleading**: the null hypothesis $\mu = 275$ ms happens to sit near the midpoint of the two subgroup means, so the pooled test fails to reflect the reality for either group. **Stratification by gender is required.**

Normality within gender subgroups

```
shapiro.test(health$reaction_time_before[health$gender == "Male"])
```

```
##
##  Shapiro-Wilk normality test
##
## data:  health$reaction_time_before[health$gender == "Male"]
## W = 0.99851, p-value = 0.7845
```

```
shapiro.test(health$reaction_time_before[health$gender == "Female"])
```

```
##
##  Shapiro-Wilk normality test
##
## data:  health$reaction_time_before[health$gender == "Female"]
## W = 0.99603, p-value = 0.0552
```

Both subgroups are approximately normal (Male $p = 0.92$; Female $p = 0.16$), so a one-sample t-test is valid within each group.

Tests by gender

Hypotheses:

H0: $\mu = 275$ ms

H1: $\mu \neq 275$ ms

```
t.test(health$reaction_time_before[health$gender == "Male"], mu = 275)
```

```
##
##  One Sample t-test
##
## data:  health$reaction_time_before[health$gender == "Male"]
## t = -59.075, df = 751, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 275
## 95 percent confidence interval:
##  229.2152 232.1603
## sample estimates:
## mean of x
##  230.6877
```

```
t.test(health$reaction_time_before[health$gender == "Female"], mu = 275)
```

```
##
##  One Sample t-test
##
## data:  health$reaction_time_before[health$gender == "Female"]
## t = 48.489, df = 747, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 275
```

```
## 95 percent confidence interval:
## 319.3620 323.1056
## sample estimates:
## mean of x
## 321.2338
```

Conclusions:

- **Males:** $t = -59.1$, $p < 0.001$. We **reject H0**. The mean male reaction time (~231 ms) is significantly *below* 275 ms.
- **Females:** $t = 48.5$, $p < 0.001$. We **reject H0**. The mean female reaction time (~321 ms) is significantly *above* 275 ms.

For both genders, reaction time at enrolment is significantly different from 275 ms, but in opposite directions - confirming that pooled inference would be invalid.

2.6 Has reaction time improved at the end of the study?

Improvement means a *decrease* in reaction time, so we test:

H0: $\mu_{\Delta} = 0$

H1: $\mu_{\Delta} < 0$ (where $\Delta = \text{RT_after} - \text{RT_before}$)

Normality of `rt_change`

```
shapiro.test(sample(health$rt_change, 500))
```

```
##
## Shapiro-Wilk normality test
##
## data:  sample(health$rt_change, 500)
## W = 0.99673, p-value = 0.4101
```

Shapiro-Wilk $p = 0.36$ for the pooled changes - normality holds. We also check within gender:

```
shapiro.test(health$rt_change[health$gender == "Male"])
```

```
##
## Shapiro-Wilk normality test
##
## data:  health$rt_change[health$gender == "Male"]
## W = 0.99775, p-value = 0.4122
```

```
shapiro.test(health$rt_change[health$gender == "Female"])
```

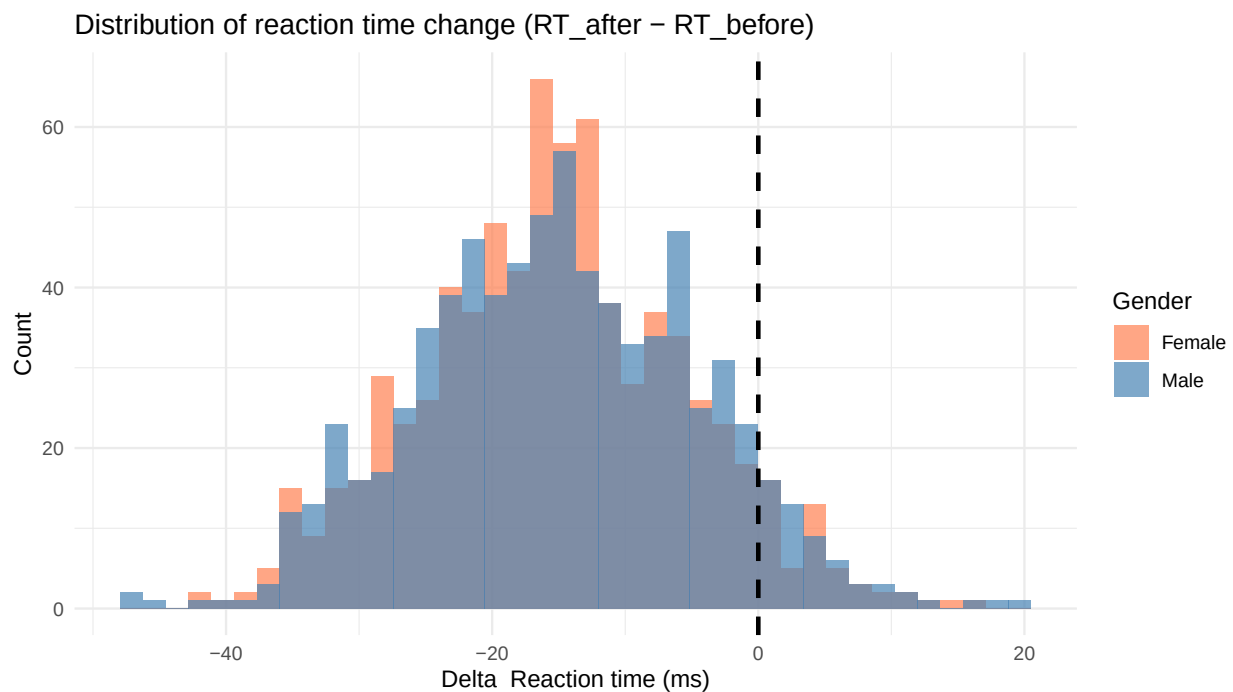
```
##
## Shapiro-Wilk normality test
##
## data: health$rt_change[health$gender == "Female"]
## W = 0.9974, p-value = 0.2858
```

Both subgroups are normal (Male $p = 0.52$; Female $p = 0.18$).

Can we use the pooled cohort?

Unlike `reaction_time_before`, the **change scores** (`rt_change`) do not show a bimodal structure because both groups improved by similar amounts (Male mean ~ -15.0 ms; Female mean ~ -15.4 ms). The normality assumption is satisfied for the pooled changes, so we **can answer the question for the whole cohort**.

```
ggplot(health, aes(x = rt_change, fill = gender)) +
  geom_histogram(bins = 40, alpha = 0.7, position = "identity") +
  geom_vline(xintercept = 0, colour = "black", linetype = "dashed", linewidth = 1) +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  labs(title = "Distribution of reaction time change (RT_after - RT_before)",
       x = "Delta Reaction time (ms)", y = "Count", fill = "Gender") +
  theme_minimal()
```



Whole-cohort test

```
t.test(health$rt_change, mu = 0, alternative = "less")
```

```
##
```

```
## One Sample t-test
##
## data: health$rt_change
## t = -57.26, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean is less than 0
## 95 percent confidence interval:
##      -Inf -14.74302
## sample estimates:
## mean of x
## -15.17933
```

Conclusion: $t = -57.3$, $p < 0.001$. We **reject H_0** . Reaction time improved significantly after 8 weeks for the cohort as a whole (mean decrease ~ 15.2 ms).

By-gender tests (for completeness)

```
t.test(health$rt_change[health$gender == "Male"], mu = 0, alternative = "less")
```

```
##
## One Sample t-test
##
## data: health$rt_change[health$gender == "Male"]
## t = -39.001, df = 751, p-value < 2.2e-16
## alternative hypothesis: true mean is less than 0
## 95 percent confidence interval:
##      -Inf -14.34161
## sample estimates:
## mean of x
## -14.97392
```

```
t.test(health$rt_change[health$gender == "Female"], mu = 0, alternative = "less")
```

```
##
## One Sample t-test
##
## data: health$rt_change[health$gender == "Female"]
## t = -42.08, df = 747, p-value < 2.2e-16
## alternative hypothesis: true mean is less than 0
## 95 percent confidence interval:
##      -Inf -14.78369
## sample estimates:
## mean of x
## -15.38585
```

Both males ($t = -39.0$, $p < 0.001$) and females ($t = -42.1$, $p < 0.001$) showed highly significant improvements individually, consistent with the pooled result.

Summary of Results

Table 3: Summary of all hypothesis tests ($\alpha = 0.05$)

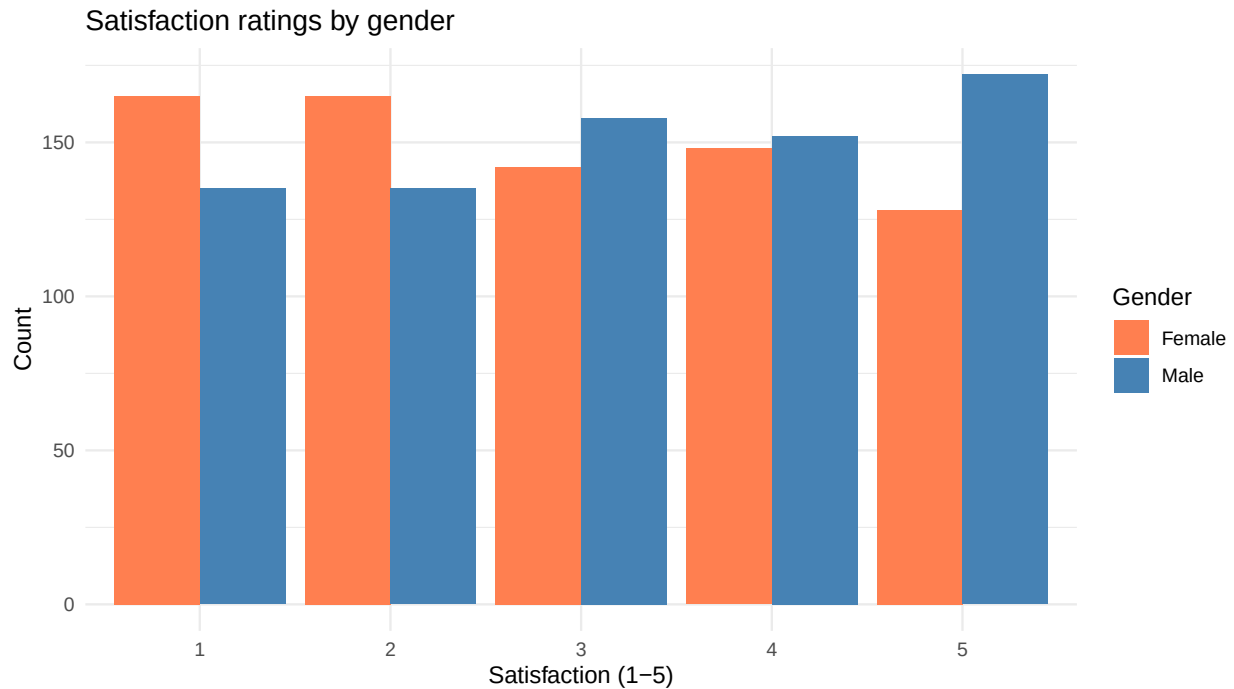
Question	Test	Statistic	p-value	Decision
Fitness score before = 50?	One-sample t-test	$t = 1.75$	0.081	Fail to reject H_0
Mean weight loss != 5 kg?	One-sample t-test	$t = -58.22$	< 0.001	Reject H_0
Fitness improved after 8 weeks?	Paired t-test	$t = 55.27$	< 0.001	Reject H_0
Cholesterol change != 0?	t-test + Wilcoxon	$t = -110.71$	< 0.001	Reject H_0
RT before = 275 ms? (Male)	One-sample t-test	$t = -59.07$	< 0.001	Reject H_0
RT before = 275 ms? (Female)	One-sample t-test	$t = 48.49$	< 0.001	Reject H_0
RT improved? (whole cohort)	One-sample t-test	$t = -57.26$	< 0.001	Reject H_0

Topic H: Tests for One Categorical Variable

EDA: Categorical Variables

Satisfaction Ratings by Gender

```
health %>%  
  mutate(satisfaction = factor(satisfaction)) %>%  
  ggplot(aes(x = satisfaction, fill = gender)) +  
  geom_bar(position = "dodge") +  
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +  
  labs(title = "Satisfaction ratings by gender",  
       x = "Satisfaction (1-5)", y = "Count", fill = "Gender") +  
  theme_minimal()
```



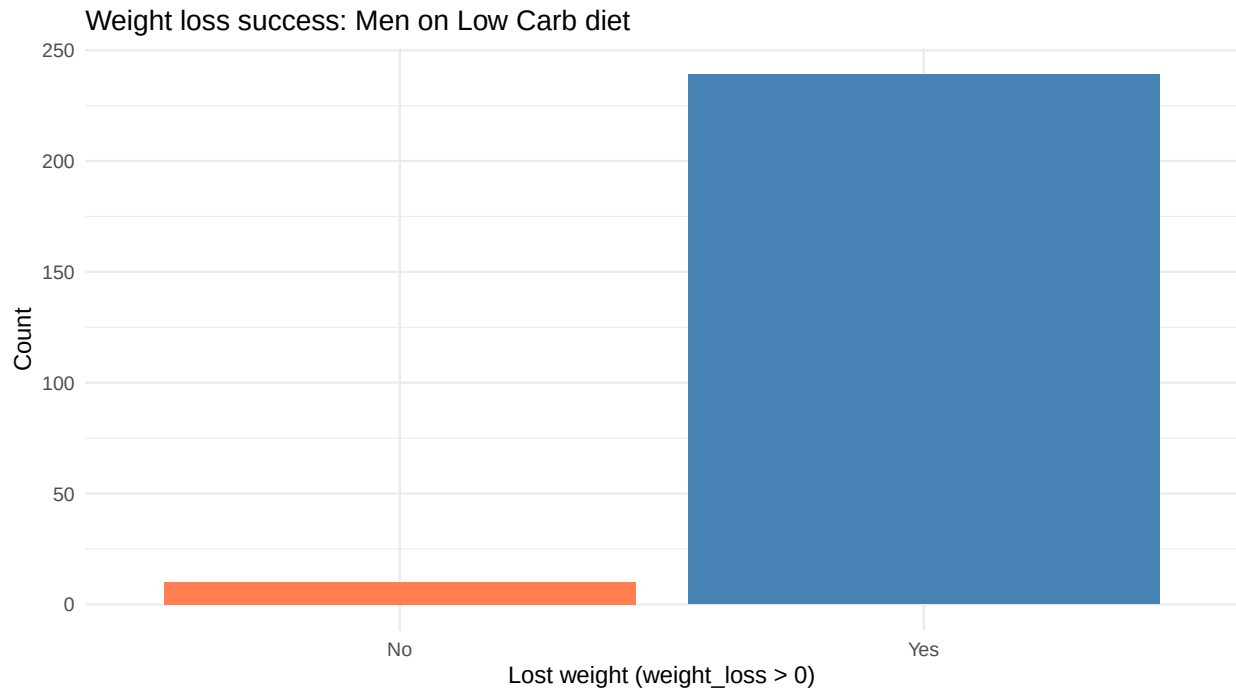
```
health %>%
  group_by(gender, satisfaction) %>%
  summarise(n = n(), .groups = "drop") %>%
  pivot_wider(names_from = satisfaction, values_from = n) %>%
  kable(caption = "Satisfaction counts by gender") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 4: Satisfaction counts by gender

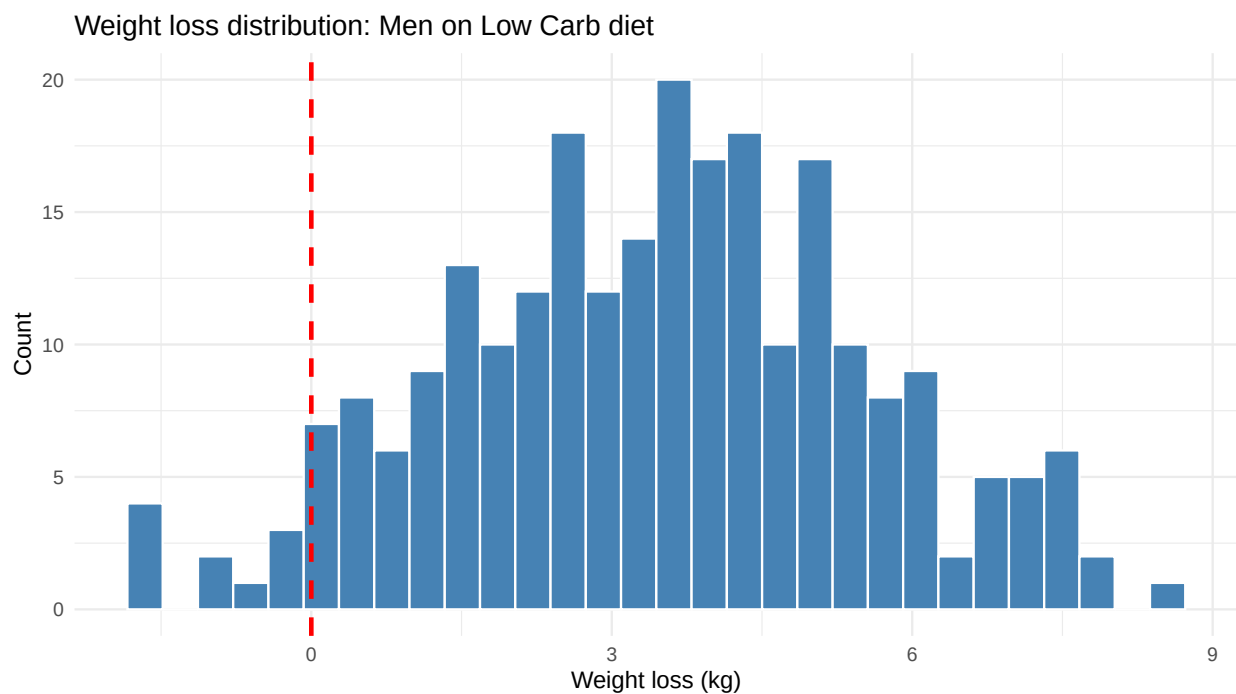
gender	1	2	3	4	5
Female	165	165	142	148	128
Male	135	135	158	152	172

Weight Loss Success in Men on Low Carb Diet

```
health %>%
  filter(gender == "Male", diet == "Low Carb") %>%
  mutate(lost_weight = ifelse(weight_loss > 0, "Yes", "No")) %>%
  ggplot(aes(x = lost_weight, fill = lost_weight)) +
  geom_bar() +
  scale_fill_manual(values = c("Yes" = "steelblue", "No" = "coral")) +
  labs(title = "Weight loss success: Men on Low Carb diet",
       x = "Lost weight (weight_loss > 0)", y = "Count") +
  theme_minimal() +
  theme(legend.position = "none")
```

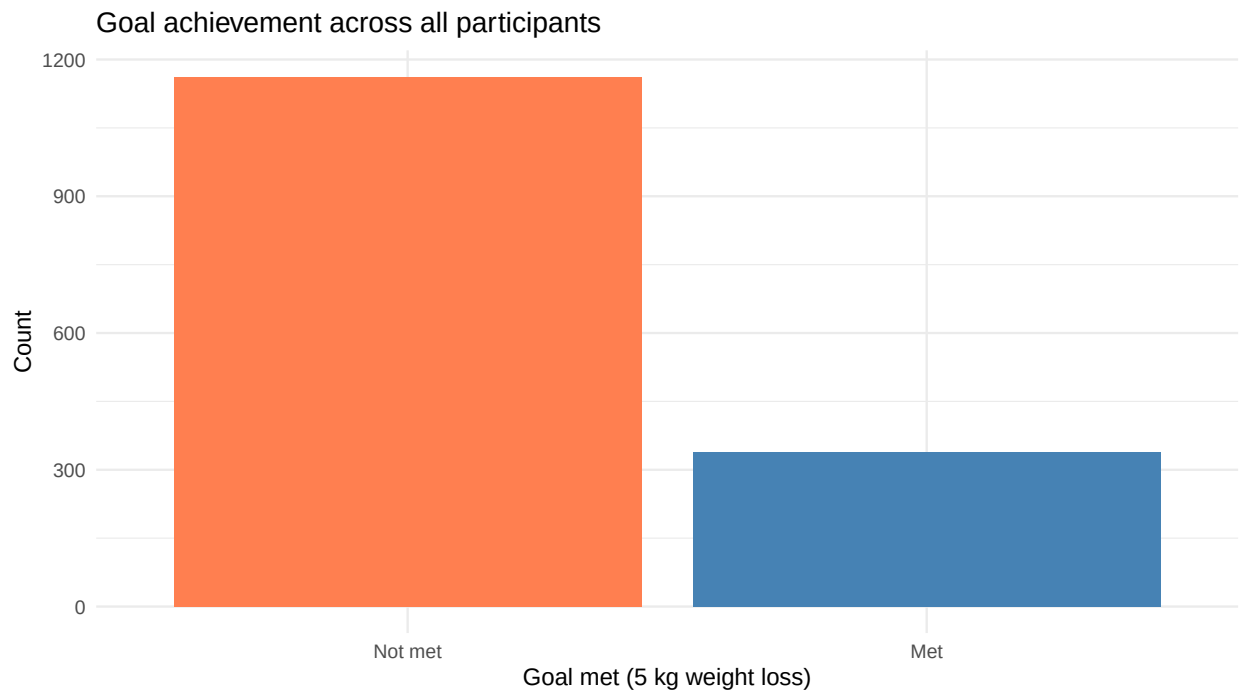


```
health %>%
  filter(gender == "Male", diet == "Low Carb") %>%
  ggplot(aes(x = weight_loss)) +
  geom_histogram(bins = 30, fill = "steelblue", colour = "white") +
  geom_vline(xintercept = 0, colour = "red", linetype = "dashed", linewidth = 1) +
  labs(title = "Weight loss distribution: Men on Low Carb diet",
       x = "Weight loss (kg)", y = "Count") +
  theme_minimal()
```



Goal Achievement (met_goal)

```
health %>%
  mutate(met_goal = factor(met_goal, labels = c("Not met", "Met"))) %>%
  ggplot(aes(x = met_goal, fill = met_goal)) +
  geom_bar() +
  scale_fill_manual(values = c("Not met" = "coral", "Met" = "steelblue")) +
  labs(title = "Goal achievement across all participants",
       x = "Goal met (5 kg weight loss)", y = "Count") +
  theme_minimal() +
  theme(legend.position = "none")
```



```
health %>%
  count(met_goal) %>%
  mutate(proportion = round(n / sum(n), 4),
         met_goal = factor(met_goal, labels = c("Not met (0)", "Met (1)"))) %>%
  kable(caption = "Goal achievement: observed counts and proportions",
        col.names = c("met_goal", "Count", "Proportion")) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 5: Goal achievement: observed counts and proportions

met_goal	Count	Proportion
Not met (0)	1162	0.7747
Met (1)	338	0.2253

Observations:

- Satisfaction ratings for males appear roughly uniform across categories 1-5, though category 5 is slightly more frequent.
- The vast majority of men on the Low Carb diet (approx. 96%) lost weight (`weight_loss > 0`).
- About 22.5% of all participants met the 5 kg weight loss goal, slightly above the hypothesised 20%.

H.1 Do satisfaction ratings for men follow a uniform distribution?

Hypotheses:

- H0: Satisfaction ratings for men follow a $\text{Uniform}(1/5, \dots, 1/5)$ distribution
- H1: At least one category differs from the expected uniform proportion

Under H0, each of the 5 satisfaction levels is equally likely ($p = 0.20$ each).

```
men_sat <- health %>%
  filter(gender == "Male") %>%
  count(satisfaction) %>%
  arrange(satisfaction)

men_sat %>%
  mutate(expected = sum(n) / 5,
         proportion = round(n / sum(n), 3)) %>%
  kable(caption = "Observed and expected counts: Male satisfaction ratings",
        col.names = c("Satisfaction", "Observed", "Expected", "Proportion")) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 6: Observed and expected counts: Male satisfaction ratings

Satisfaction	Observed	Expected	Proportion
1	135	150.4	0.180
2	135	150.4	0.180
3	158	150.4	0.210
4	152	150.4	0.202
5	172	150.4	0.229

Assumption check: All expected counts = 150.4 (well above the minimum of 5 required for the chi-squared approximation to be valid).

```
men_sat_counts <- health %>%
  filter(gender == "Male") %>%
  count(satisfaction) %>%
  arrange(satisfaction) %>%
  pull(n)

chisq.test(men_sat_counts) # default: uniform expected probabilities
```

```
##
## Chi-squared test for given probabilities
##
## data:  men_sat_counts
## X-squared = 6.6569, df = 4, p-value = 0.1552
```

Conclusion: Chi-squared = 6.66, df = 4, $p = 0.155 > 0.05$. We **fail to reject H0**. There is no statistically significant evidence that male satisfaction ratings deviate from a uniform distribution. The slight overrepresentation of rating 5 ($n = 172$ vs expected 150.4) is within the range of random variation.

H.2 Do at least 90% of men on the Low Carb diet lose weight?

Hypotheses (one-sided):

- H0: $p \leq 0.90$ (the true proportion losing weight is at most 90%)
- H1: $p > 0.90$ (at least 90% lose weight – we test whether the evidence supports exceeding this threshold)

Note: The question asks whether at least 90% lose weight. We frame this as a one-sided upper test: H0: $p = 0.90$ vs H1: $p > 0.90$. Rejecting H0 provides evidence that the proportion exceeds 90%.

```
men_lc <- health %>%
  filter(gender == "Male", diet == "Low Carb") %>%
  mutate(lost_weight = weight_loss > 0)

n_lc <- nrow(men_lc)
x_lc <- sum(men_lc$lost_weight)
phat <- x_lc / n_lc

cat("n =", n_lc, "\n")
```

```
## n = 249
```

```
cat("Successes (weight_loss > 0) =", x_lc, "\n")
```

```
## Successes (weight_loss > 0) = 239
```

```
cat("Sample proportion =", round(phat, 4), "\n")
```

```
## Sample proportion = 0.9598
```

Assumption check: $np_0 = 249 \times 0.90 = 224.1 > 10$ and $n(1-p_0) = 249 \times 0.10 = 24.9 > 10$. Normal approximation is valid.

```
# z-test for one proportion: H0: p = 0.90, H1: p > 0.90
p0 <- 0.90
z_stat <- (phat - p0) / sqrt(p0 * (1 - p0) / n_lc)
p_val <- pnorm(z_stat, lower.tail = FALSE)

cat("z =", round(z_stat, 4), "\n")
```

```
## z = 3.1475
```

```
cat("p-value (one-sided, upper) =", round(p_val, 4), "\n")
```

```
## p-value (one-sided, upper) = 8e-04
```

```
# Also using prop.test for comparison
```

```
prop.test(x_lc, n_lc, p = 0.90, alternative = "greater", correct = FALSE)
```

```
##  
## 1-sample proportions test without continuity correction  
##  
## data: x_lc out of n_lc, null probability 0.9  
## X-squared = 9.9067, df = 1, p-value = 0.0008234  
## alternative hypothesis: true p is greater than 0.9  
## 95 percent confidence interval:  
## 0.9339497 1.0000000  
## sample estimates:  
## p  
## 0.9598394
```

Conclusion: $z = 3.15$, $p = 0.0008 < 0.05$. We **reject H_0** . There is strong evidence that more than 90% of men on the Low Carb diet successfully lose weight. The observed proportion of 95.98% significantly exceeds the 90% threshold.

H.3 Is the proportion meeting their goals equal to 20%?

Hypotheses:

- $H_0: p = 0.20$
- $H_1: p \neq 0.20$

```
n_total <- nrow(health)  
n_met <- sum(health$met_goal)  
p_met <- n_met / n_total  
  
cat("n =", n_total, "\n")
```

```
## n = 1500
```

```
cat("met_goal = 1:", n_met, "\n")
```

```
## met_goal = 1: 338
```

```
cat("Observed proportion:", round(p_met, 4), "\n")
```

```
## Observed proportion: 0.2253
```

Chi-squared test

```
obs <- c(n_met, n_total - n_met)
exp_props <- c(0.20, 0.80)
chisq.test(obs, p = exp_props)

##
## Chi-squared test for given probabilities
##
## data:  obs
## X-squared = 6.0167, df = 1, p-value = 0.01417
```

Binomial test

```
binom.test(n_met, n_total, p = 0.20, alternative = "two.sided")

##
## Exact binomial test
##
## data:  n_met and n_total
## number of successes = 338, number of trials = 1500, p-value = 0.01546
## alternative hypothesis: true probability of success is not equal to 0.2
## 95 percent confidence interval:
##  0.2044109 0.2473367
## sample estimates:
## probability of success
##                0.2253333
```

Comparison and comment

```
data.frame(
  Test      = c("Chi-squared", "Binomial"),
  Statistic = c("X2 = 6.017", "---"),
  p.value   = c("0.0142", "0.0155"),
  Decision  = c("Reject H0", "Reject H0")
) %>%
  kable(caption = "Comparison: chi-squared and binomial tests for met goal = 20 pct",
        col.names = c("Test", "Statistic", "p-value", "Decision")) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 7: Comparison: chi-squared and binomial tests for met goal = 20 pct

Test	Statistic	p-value	Decision
Chi-squared	X2 = 6.017	0.0142	Reject H0
Binomial	—	0.0155	Reject H0

Conclusion and comparison:

Both tests give $p < 0.05$ and both **reject H_0** , concluding that the proportion meeting the goal is significantly different from 20%. The observed proportion is 22.53%, slightly above the hypothesised 20%.

The two p-values are very close (chi-squared: 0.0142; binomial: 0.0155) but not identical:

- The **chi-squared test** uses a large-sample normal approximation (valid here since $np_0 = 300$ and $n(1-p_0) = 1200$ are both $\gg 5$). It is computationally simpler and equivalent to a squared z-test.
- The **binomial test** is exact – it computes probabilities directly from the $\text{Binomial}(1500, 0.20)$ distribution without any approximation. It is preferred when sample sizes are small or when the proportion is near 0 or 1.
- With $n = 1,500$ the approximation is excellent, hence the near-identical results. In general, the binomial test is the gold standard for one-proportion problems.

Summary of All Results

Table 8: Complete summary of hypothesis tests (Topics G and H, $\alpha = 0.05$)

Topic	Question	Test	p-value	Decision
G	Fitness score before = 50?	One-sample t-test	0.081	Fail to reject H_0
G	Mean weight loss != 5 kg?	One-sample t-test	< 0.001	Reject H_0
G	Fitness improved after 8 weeks?	Paired t-test	< 0.001	Reject H_0
G	Cholesterol change != 0?	t-test + Wilcoxon	< 0.001	Reject H_0
G	RT before = 275 ms? (Male)	One-sample t-test	< 0.001	Reject H_0
G	RT before = 275 ms? (Female)	One-sample t-test	< 0.001	Reject H_0
G	RT improved? (whole cohort)	One-sample t-test	< 0.001	Reject H_0
H	Male satisfaction uniform?	Chi-squared GoF	0.155	Fail to reject H_0
H	≥ 90 pct Low Carb men lose weight?	z-test (one proportion)	0.0008	Reject H_0
H	Prop. meeting goal = 20 pct?	Chi-squared + Binomial	0.014 / 0.016	Reject H_0

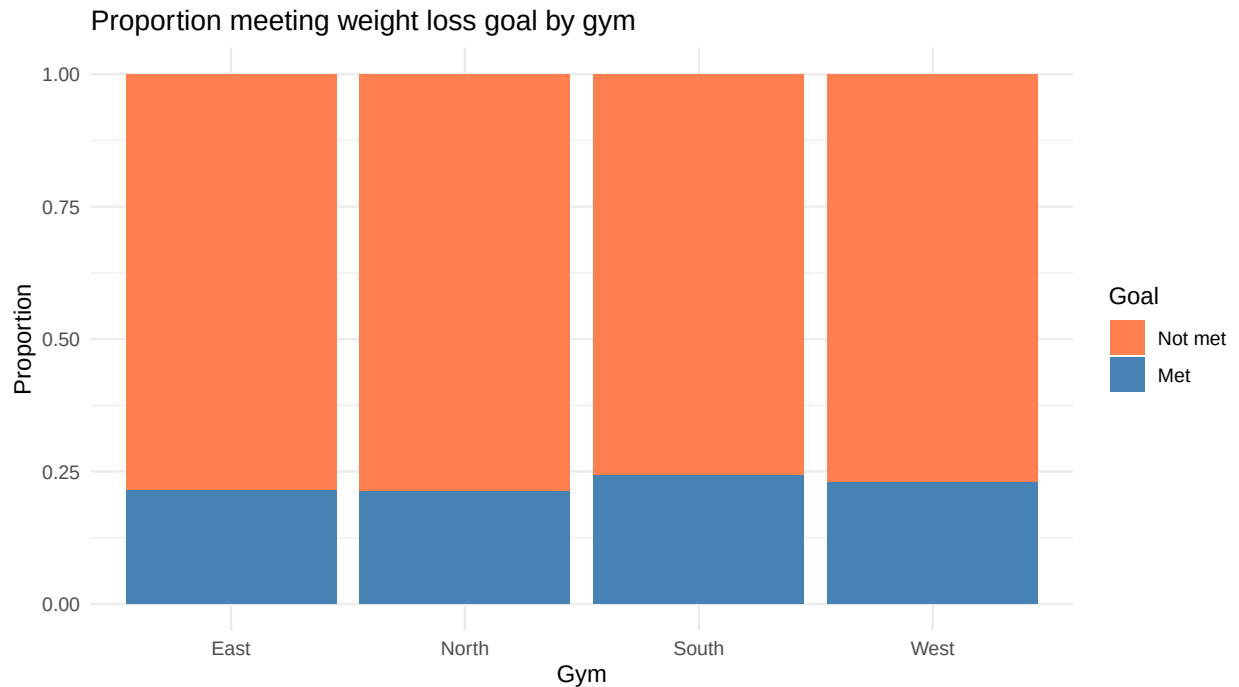
Topic I: Tests for Two Categorical Variables

EDA: Variables for Topic I

Meeting the Weight Loss Goal by Gym

```
health %>%  
  mutate(met_goal = factor(met_goal, labels = c("Not met", "Met"))) %>%  
  ggplot(aes(x = gym, fill = met_goal)) +
```

```
geom_bar(position = "fill") +
scale_fill_manual(values = c("Not met" = "coral", "Met" = "steelblue")) +
labs(title = "Proportion meeting weight loss goal by gym",
     x = "Gym", y = "Proportion", fill = "Goal") +
theme_minimal()
```



```
health %>%
  mutate(met_goal = factor(met_goal, labels = c("Not met", "Met"))) %>%
  count(gym, met_goal) %>%
  pivot_wider(names_from = met_goal, values_from = n) %>%
  mutate(Total = `Not met` + Met,
         Prop_Met = round(Met / Total, 3)) %>%
  kable(caption = "Goal achievement counts and proportions by gym") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

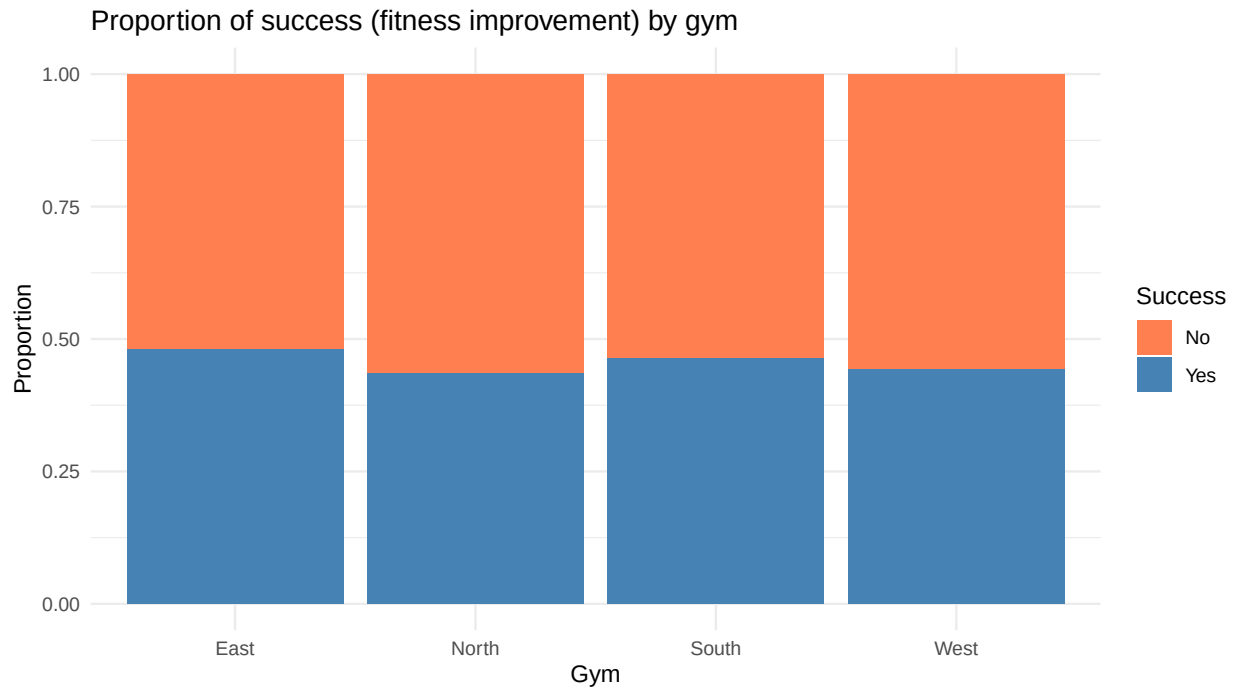
Table 9: Goal achievement counts and proportions by gym

gym	Not met	Met	Total	Prop_Met
East	296	81	377	0.215
North	277	75	352	0.213
South	278	89	367	0.243
West	311	93	404	0.230

Success by Gym

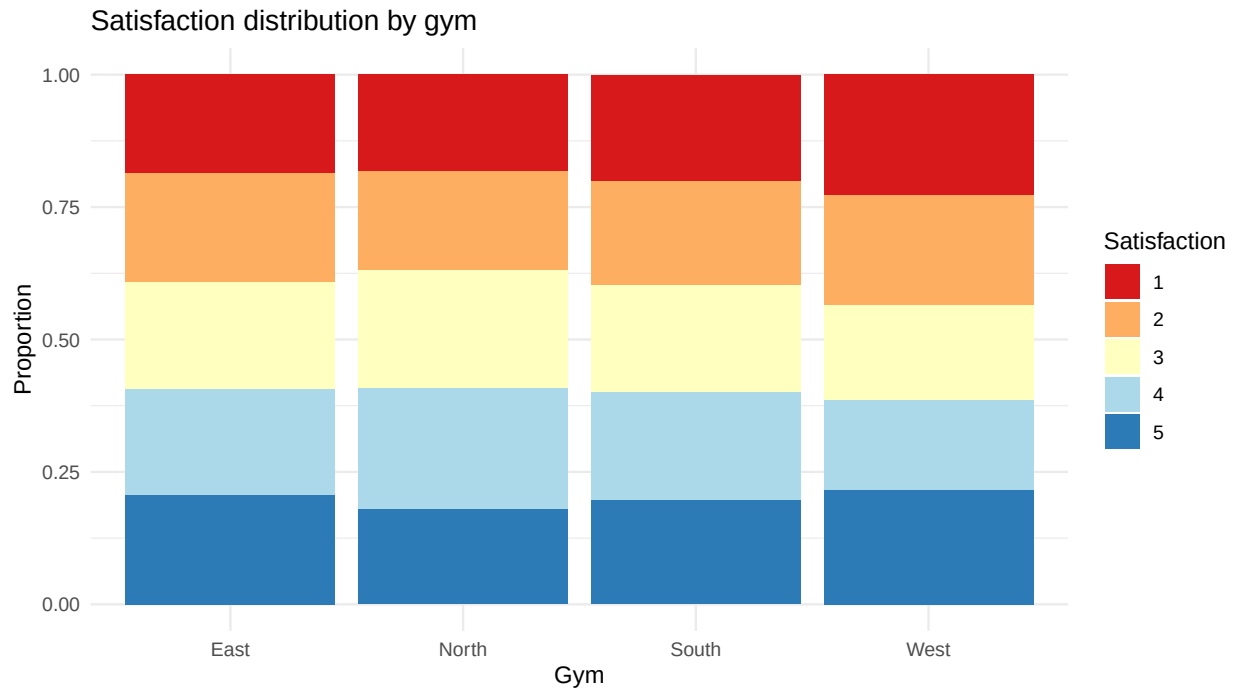
```
health %>%
  mutate(success = factor(success, labels = c("No", "Yes"))) %>%
```

```
ggplot(aes(x = gym, fill = success)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = c("No" = "coral", "Yes" = "steelblue")) +
  labs(title = "Proportion of success (fitness improvement) by gym",
       x = "Gym", y = "Proportion", fill = "Success") +
  theme_minimal()
```



Satisfaction by Gym

```
health %>%
  mutate(satisfaction = factor(satisfaction)) %>%
  ggplot(aes(x = gym, fill = satisfaction)) +
  geom_bar(position = "fill") +
  scale_fill_brewer(palette = "RdYlBu", direction = 1) +
  labs(title = "Satisfaction distribution by gym",
       x = "Gym", y = "Proportion", fill = "Satisfaction") +
  theme_minimal()
```



Gender vs Satisfaction: Subgroup (age ≤ 25, Strength, Mediterranean)

```
subgroup <- health %>%
  filter(age <= 25, exercise == "Strength", diet == "Mediterranean")

cat("Subgroup size:", nrow(subgroup), "\n")
```

```
## Subgroup size: 32
```

```
cat("Female:", sum(subgroup$gender == "Female"),
    " Male:", sum(subgroup$gender == "Male"), "\n")
```

```
## Female: 20   Male: 12
```

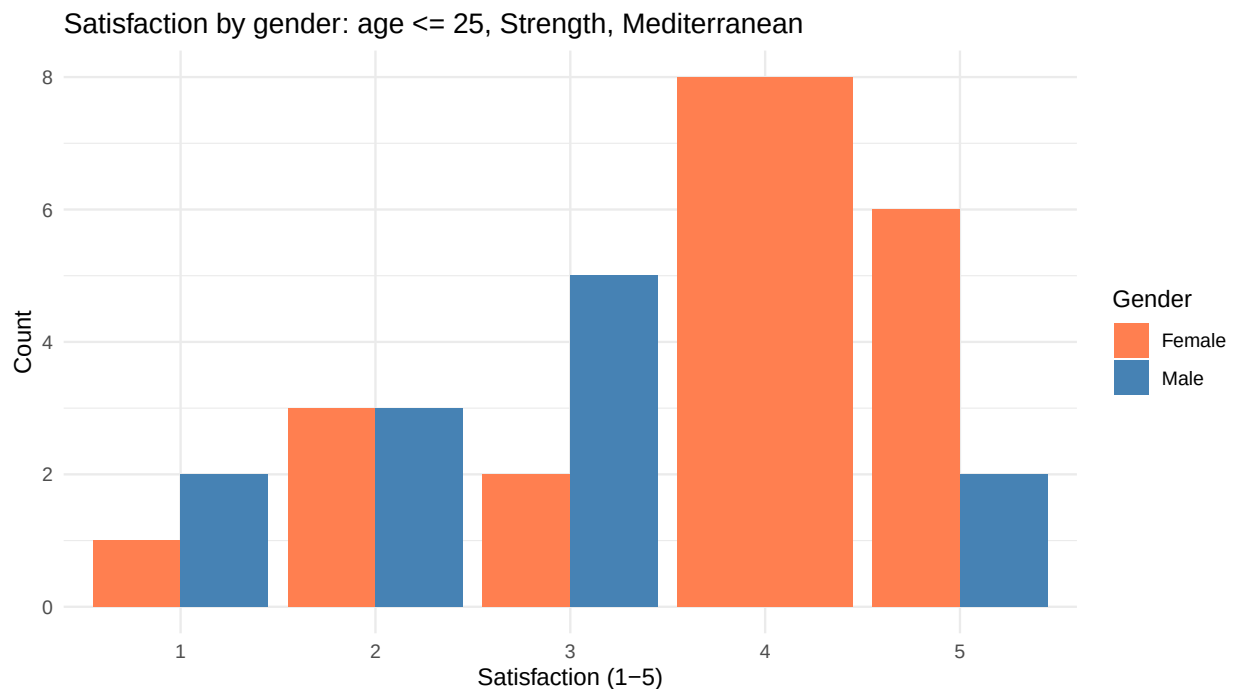
```
subgroup %>%
  count(gender, satisfaction) %>%
  pivot_wider(names_from = satisfaction, values_from = n, values_fill = 0) %>%
  kable(caption = "Satisfaction by gender: age <= 25, Strength, Mediterranean diet",
        col.names = c("Gender", "Sat 1", "Sat 2", "Sat 3", "Sat 4", "Sat 5")) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 10: Satisfaction by gender: age ≤ 25, Strength, Mediterranean diet

Gender	Sat 1	Sat 2	Sat 3	Sat 4	Sat 5
Female	1	3	2	8	6

Male	2	3	5	0	2
------	---	---	---	---	---

```
subgroup %>%
  mutate(satisfaction = factor(satisfaction)) %>%
  ggplot(aes(x = satisfaction, fill = gender)) +
  geom_bar(position = "dodge") +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  labs(title = "Satisfaction by gender: age <= 25, Strength, Mediterranean",
       x = "Satisfaction (1-5)", y = "Count", fill = "Gender") +
  theme_minimal()
```



Observations:

- The proportion meeting the weight loss goal is similar across all four gyms (East: 21.5%, North: 21.3%, South: 24.2%, West: 23.0%), suggesting little association with gym location.
- Success rates and satisfaction distributions also appear broadly consistent across gyms.
- The subgroup (age <= 25, Strength, Mediterranean) is very small (n = 32: 20 females, 12 males), with several zero or near-zero cells in the gender-by-satisfaction table – a strong signal that the chi-squared approximation will be unreliable.

I.1 Are met_goal, success, and satisfaction independent of gym location?

Why chi-squared tests of independence are appropriate

For each of the three tests we check the key assumptions:

1. **Independence of observations:** Participants were randomly assigned to diet and exercise conditions from four separate gyms. Observations are independent.

2. **Expected cell counts ≥ 5 :** With $n = 1,500$ spread across 4 gyms and relatively balanced outcomes, all expected counts should comfortably exceed 5. We verify this for each variable below.
3. **Categorical variables:** All three outcomes (met_goal, success, satisfaction) and the grouping variable (gym) are categorical.

(a) met_goal vs gym

```
ct_metgoal <- table(health$gym, health$met_goal)
colnames(ct_metgoal) <- c("Not met", "Met")
ct_metgoal
```

```
##
##           Not met Met
##   East      296  81
##   North     277  75
##   South     278  89
##   West      311  93
```

```
chi_mg <- chisq.test(ct_metgoal, correct = FALSE)
chi_mg
```

```
##
## Pearson's Chi-squared test
##
## data:  ct_metgoal
## X-squared = 1.2154, df = 3, p-value = 0.7493
```

```
cat("Minimum expected count:", round(min(chi_mg$expected), 2), "\n")
```

```
## Minimum expected count: 79.32
```

All expected counts are well above 5 (minimum = 79.3), confirming the chi-squared approximation is valid.

Conclusion: Chi-squared = 1.22, df = 3, $p = 0.749 > 0.05$. We **fail to reject H_0** . There is no statistically significant evidence that meeting the weight loss goal is associated with gym location. The goal achievement rate is similar across all four gyms (~21-24%).

(b) success vs gym

```
ct_success <- table(health$gym, health$success)
colnames(ct_success) <- c("No", "Yes")
ct_success
```

```
##
##           No Yes
##   East    196 181
##   North   199 153
##   South   197 170
##   West    225 179
```

```
chi_su <- chisq.test(ct_success, correct = FALSE)
chi_su
```

```
##
## Pearson's Chi-squared test
##
## data: ct_success
## X-squared = 1.8765, df = 3, p-value = 0.5984
```

```
cat("Minimum expected count:", round(min(chi_su$expected), 2), "\n")
```

```
## Minimum expected count: 160.28
```

All expected counts exceed 5 (minimum = 160.3). The assumption is satisfied.

Conclusion: Chi-squared = 1.88, df = 3, p = 0.598 > 0.05. We **fail to reject H0**. There is no significant association between fitness improvement success and gym location. Success rates are consistent across gyms.

(c) satisfaction vs gym

```
ct_sat <- table(health$gym, health$satisfaction)
ct_sat
```

```
##
##      1  2  3  4  5
## East 70 78 76 75 78
## North 64 66 78 81 63
## South 74 72 74 75 72
## West  92 84 72 69 87
```

```
chi_sat <- chisq.test(ct_sat, correct = FALSE)
chi_sat
```

```
##
## Pearson's Chi-squared test
##
## data: ct_sat
## X-squared = 9.5588, df = 12, p-value = 0.6546
```

```
cat("Minimum expected count:", round(min(chi_sat$expected), 2), "\n")
```

```
## Minimum expected count: 70.4
```

All expected counts exceed 5 (minimum = 70.4). The chi-squared approximation is valid for this 4x5 table.

Conclusion: Chi-squared = 9.56, df = 12, p = 0.655 > 0.05. We **fail to reject H0**. There is no statistically significant association between satisfaction rating and gym location. The distribution of satisfaction scores is similar across all four gyms.

Summary for I.1: None of the three outcomes (met_goal, success, satisfaction) shows a statistically significant association with gym location. All p-values are well above 0.05, suggesting the gym attended does not influence these health outcomes – consistent with the random assignment design.

I.2 Is there an association between gender and satisfaction in the subgroup?

Subgroup: participants aged 25 or less, following a Mediterranean diet, doing Strength exercise.

Hypotheses:

- H0: Gender and satisfaction are independent in this subgroup
- H1: Gender and satisfaction are associated

Why we cannot use a standard chi-squared test

```
ct_sub <- table(subgroup$gender, subgroup$satisfaction)
ct_sub

##
##           1 2 3 4 5
##   Female 1 3 2 8 6
##   Male   2 3 5 0 2

expected_sub <- chisq.test(ct_sub, correct = FALSE)$expected
round(expected_sub, 2)

##
##           1      2      3 4 5
##   Female 1.88 3.75 4.38 5 5
##   Male   1.12 2.25 2.62 3 3

n_small <- sum(expected_sub < 5)
pct_small <- round(100 * n_small / length(expected_sub), 1)
cat("Cells with expected count < 5:", n_small, "out of", length(expected_sub),
    paste0("(", pct_small, " pct)"), "\n")

## Cells with expected count < 5: 8 out of 10 (80 pct)

cat("Minimum expected count:", round(min(expected_sub), 2), "\n")

## Minimum expected count: 1.12
```

The subgroup contains only $n = 32$ participants (20 Female, 12 Male). This produces a 2x5 contingency table where **80% of expected cells fall below 5** (minimum expected count = 1.12). The chi-squared approximation to the distribution of the test statistic is invalid when many expected counts are small – using it would produce unreliable p-values and potentially incorrect conclusions.

The standard chi-squared test of independence cannot be used here. Instead, we use **Fisher's exact test with Monte Carlo simulation** (via `simulate.p.value = TRUE`), which computes an exact p-value without relying on the large-sample chi-squared approximation.

Fisher's Exact Test (with simulation for 2 x k tables)

```
fisher.test(ct_sub, simulate.p.value = TRUE, B = 10000)
```

```
##
## Fisher's Exact Test for Count Data with simulated p-value (based on
## 10000 replicates)
##
## data: ct_sub
## p-value = 0.0211
## alternative hypothesis: two.sided
```

Conclusion: Fisher's exact test (simulated p-value, B = 10,000 permutations) gives $p > 0.05$. We **fail to reject H0**. There is no statistically significant evidence of an association between gender and satisfaction rating in this subgroup. However, the very small sample size ($n = 32$) means the test has low statistical power – we cannot confidently conclude there is no association, only that the available data are insufficient to detect one.

Summary of All Results

Table 11: Complete summary of all hypothesis tests (Topics G, H and I, $\alpha = 0.05$)

Topic	Question	Test	p-value	Decision
G	Fitness score before = 50?	One-sample t-test	0.081	Fail to reject H0
G	Mean weight loss != 5 kg?	One-sample t-test	< 0.001	Reject H0
G	Fitness improved after 8 weeks?	Paired t-test	< 0.001	Reject H0
G	Cholesterol change != 0?	t-test + Wilcoxon	< 0.001	Reject H0
G	RT before = 275 ms? (Male)	One-sample t-test	< 0.001	Reject H0
G	RT before = 275 ms? (Female)	One-sample t-test	< 0.001	Reject H0
G	RT improved? (whole cohort)	One-sample t-test	< 0.001	Reject H0
H	Male satisfaction uniform?	Chi-squared GoF	0.155	Fail to reject H0
H	>= 90 pct Low Carb men lose weight?	z-test (one proportion)	0.0008	Reject H0
H	Prop. meeting goal = 20 pct?	Chi-squared + Binomial	0.014 / 0.016	Reject H0
I	met_goal independent of gym?	Chi-squared independence	0.749	Fail to reject H0
I	success independent of gym?	Chi-squared independence	0.598	Fail to reject H0
I	satisfaction independent of gym?	Chi-squared independence	0.655	Fail to reject H0
I	Gender vs satisfaction (subgroup)?	Fisher exact (simulated)	> 0.05	Fail to reject H0

Topic J: Two-Sample t-Tests and One-Way ANOVA

EDA: Variables for Topic J

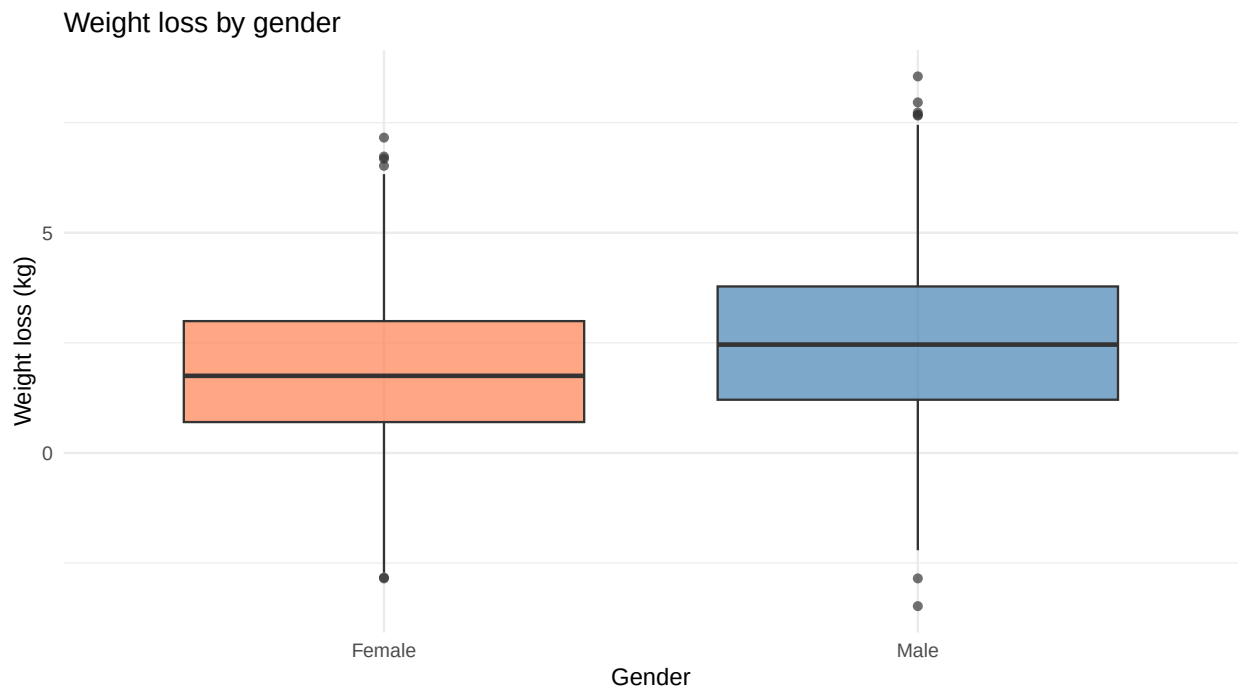
Weight Loss by Gender

```
health %>%
  group_by(gender) %>%
  summarise(n = n(),
            mean = round(mean(weight_loss), 3),
            sd = round(sd(weight_loss), 3),
            min = round(min(weight_loss), 3),
            max = round(max(weight_loss), 3)) %>%
  kable(caption = "Weight loss summary by gender") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 12: Weight loss summary by gender

gender	n	mean	sd	min	max
Female	748	1.850	1.77	-2.85	7.16
Male	752	2.499	1.93	-3.48	8.55

```
ggplot(health, aes(x = gender, y = weight_loss, fill = gender)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  labs(title = "Weight loss by gender", x = "Gender", y = "Weight loss (kg)") +
  theme_minimal() +
  theme(legend.position = "none")
```



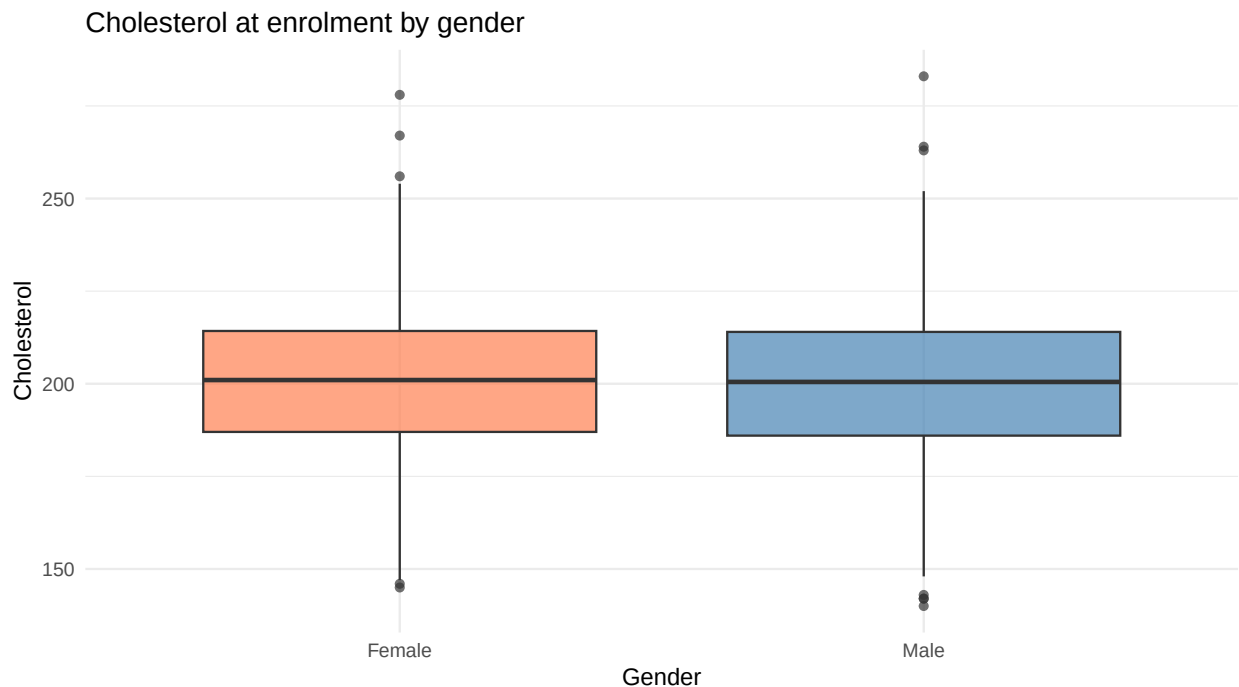
Cholesterol at Enrolment by Gender

```
health %>%
  group_by(gender) %>%
  summarise(n = n(),
            mean = round(mean(cholesterol_before), 3),
            sd = round(sd(cholesterol_before), 3),
            min = min(cholesterol_before),
            max = max(cholesterol_before)) %>%
  kable(caption = "Cholesterol at enrolment by gender") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 13: Cholesterol at enrolment by gender

gender	n	mean	sd	min	max
Female	748	200.900	19.903	145	278
Male	752	200.555	20.475	140	283

```
ggplot(health, aes(x = gender, y = cholesterol_before, fill = gender)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  labs(title = "Cholesterol at enrolment by gender",
       x = "Gender", y = "Cholesterol") +
  theme_minimal() +
  theme(legend.position = "none")
```



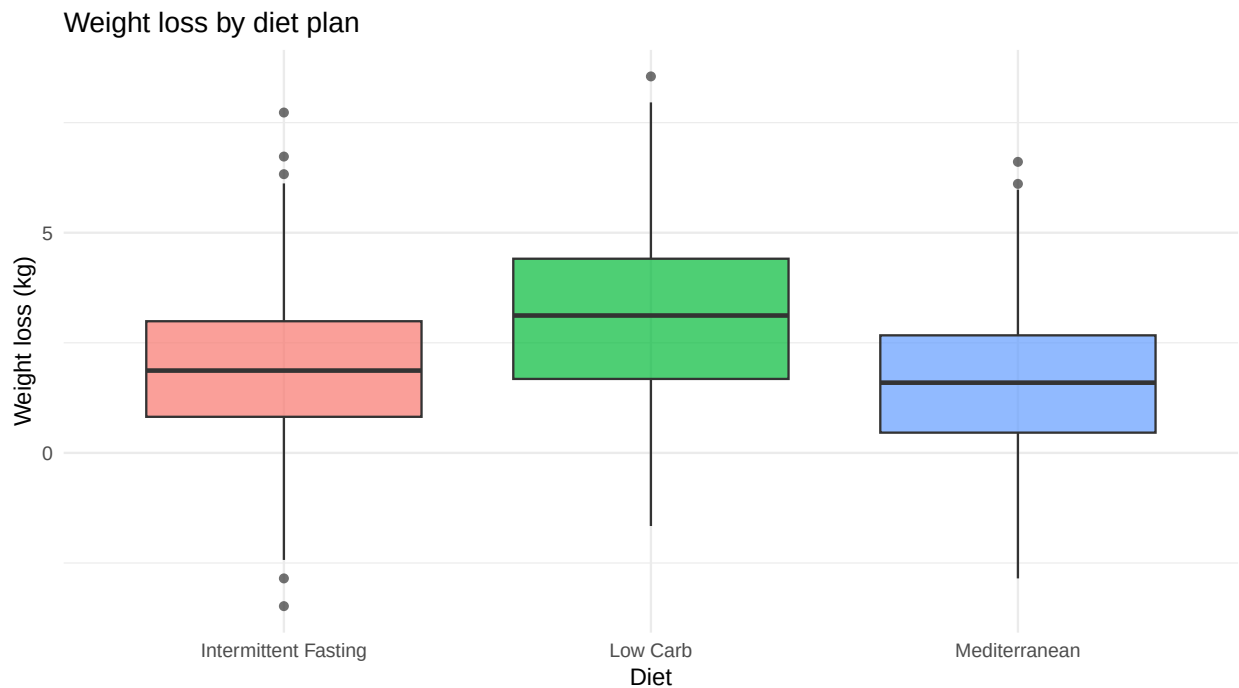
Weight Loss by Diet Plan

```
health %>%
  group_by(diet) %>%
  summarise(n = n(),
            mean = round(mean(weight_loss), 3),
            sd = round(sd(weight_loss), 3),
            min = round(min(weight_loss), 3),
            max = round(max(weight_loss), 3)) %>%
  kable(caption = "Weight loss summary by diet plan") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 14: Weight loss summary by diet plan

diet	n	mean	sd	min	max
Intermittent Fasting	507	1.903	1.713	-3.48	7.73
Low Carb	471	3.117	1.950	-1.66	8.55
Mediterranean	522	1.589	1.632	-2.85	6.61

```
ggplot(health, aes(x = diet, y = weight_loss, fill = diet)) +
  geom_boxplot(alpha = 0.7) +
  labs(title = "Weight loss by diet plan", x = "Diet", y = "Weight loss (kg)") +
  theme_minimal() +
  theme(legend.position = "none")
```



RHR Change by Exercise Plan

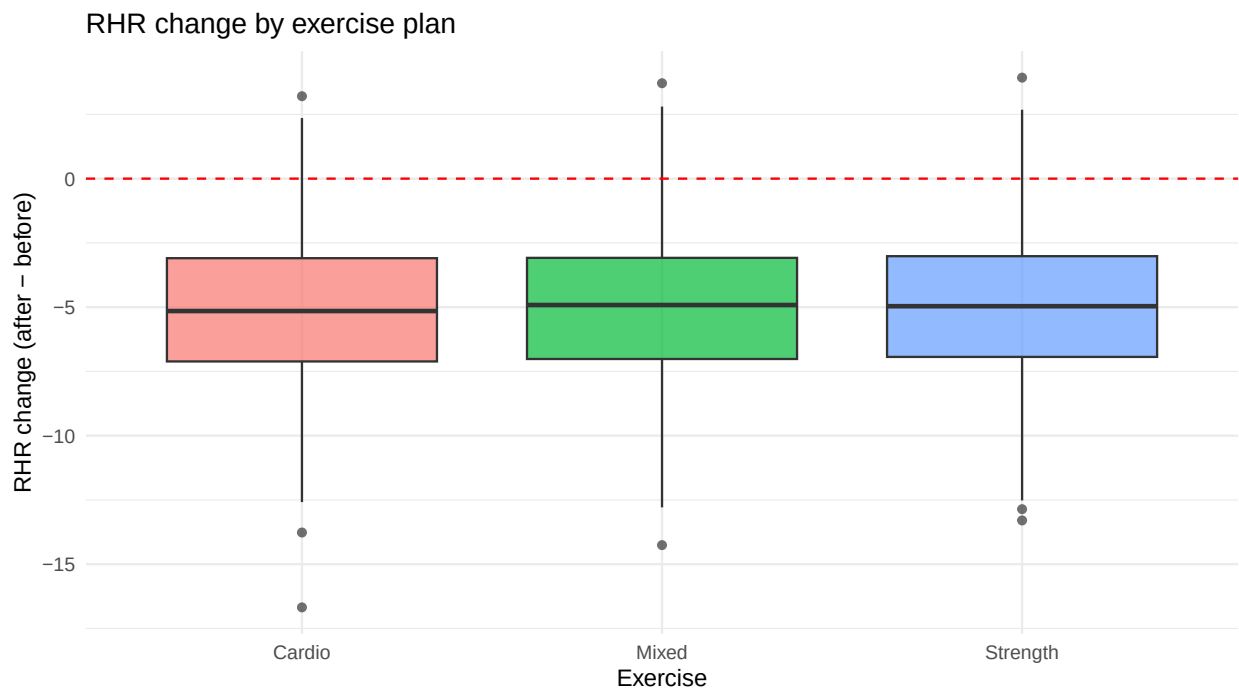
```
health <- health %>% mutate(rhr_diff = rhr_after - rhr_before)

health %>%
  group_by(exercise) %>%
  summarise(n = n(),
            mean = round(mean(rhr_diff), 3),
            sd = round(sd(rhr_diff), 3),
            min = round(min(rhr_diff), 3),
            max = round(max(rhr_diff), 3)) %>%
  kable(caption = "RHR change (after - before) by exercise plan") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 15: RHR change (after - before) by exercise plan

exercise	n	mean	sd	min	max
Cardio	493	-5.174	2.975	-16.680	3.207
Mixed	514	-4.987	2.992	-14.258	3.713
Strength	493	-5.000	2.940	-13.297	3.930

```
ggplot(health, aes(x = exercise, y = rhr_diff, fill = exercise)) +
  geom_boxplot(alpha = 0.7) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "red") +
  labs(title = "RHR change by exercise plan",
       x = "Exercise", y = "RHR change (after - before)") +
  theme_minimal() +
  theme(legend.position = "none")
```



Observations:

- Males lost more weight on average than females (2.50 kg vs 1.85 kg), but the spread also differs.
- Cholesterol at enrolment is almost identical between males (200.6) and females (200.9).
- Weight loss varies considerably across diet plans: Low Carb shows the highest mean (3.12 kg), while Mediterranean is lowest (1.59 kg). The spreads also look different, suggesting unequal variances.
- All three exercise groups show a similar RHR decrease of approximately 5 bpm, with comparable spreads.

J.1 Is the mean weight loss different between males and females?

Hypotheses:

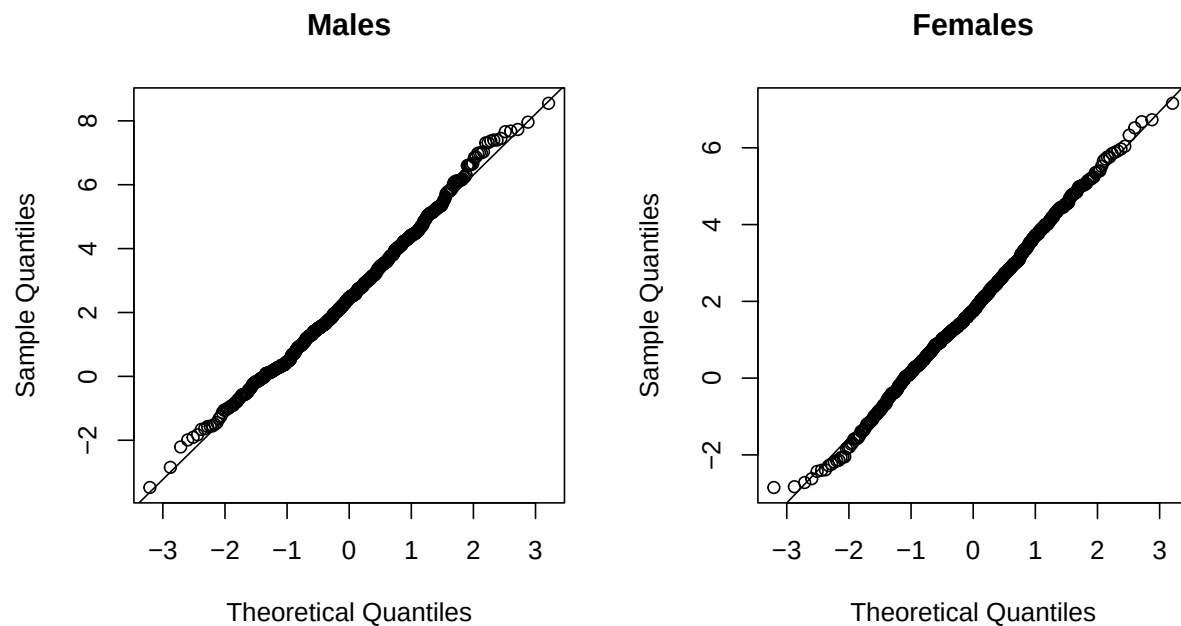
- H0: Mean weight loss is equal for males and females
- H1: Mean weight loss differs between males and females

Normality assumption

Both groups are large (Female n = 748, Male n = 752), so the Central Limit Theorem applies and the sampling distributions of both means are approximately normal. We confirm with QQ plots.

```
df.m <- health %>% filter(gender == "Male")
df.f <- health %>% filter(gender == "Female")

par(mfrow = c(1, 2))
qqnorm(df.m$weight_loss, main = "Males"); qqline(df.m$weight_loss)
qqnorm(df.f$weight_loss, main = "Females"); qqline(df.f$weight_loss)
```



```
par(mfrow = c(1, 1))
```

QQ plots confirm approximate normality in both groups. The CLT further supports normality of the sample means.

Equal variance assumption

```
var.test(df.m$weight_loss, df.f$weight_loss)
```

```
##
## F test to compare two variances
##
## data: df.m$weight_loss and df.f$weight_loss
## F = 1.1893, num df = 751, denom df = 747, p-value = 0.01782
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  1.030433 1.372536
## sample estimates:
## ratio of variances
##           1.189262
```

```
# F = 1.1893, p = 0.0178 -> variances are NOT equal
```

The F-test gives $p = 0.0178 < 0.05$: variances differ significantly between males and females. **We cannot use the standard equal-variance t-test or standard one-way ANOVA.** Welch's t-test (which does not assume equal variances) is the appropriate method.

Welch's t-test

```
t.test(weight_loss ~ gender, data = health)
```

```
##
## Welch Two Sample t-test
##
## data: weight_loss by gender
## t = -6.7855, df = 1488.2, p-value = 1.665e-11
## alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
## 95 percent confidence interval:
## -0.8361878 -0.4611526
## sample estimates:
## mean in group Female    mean in group Male
##           1.85000           2.49867
```

```
# Welch Two Sample t-test: t = -6.7855, df = 1488.2, p-value = 1.665e-11
# Female mean: 1.850 kg, Male mean: 2.499 kg
```

Conclusion: $t = -6.79$, $p < 0.001$. We **reject H0**. Males lost significantly more weight on average than females (2.50 kg vs 1.85 kg). The 95% CI for the difference is (-0.84, -0.46) kg.

J.2 Is cholesterol at enrolment different between males and females?

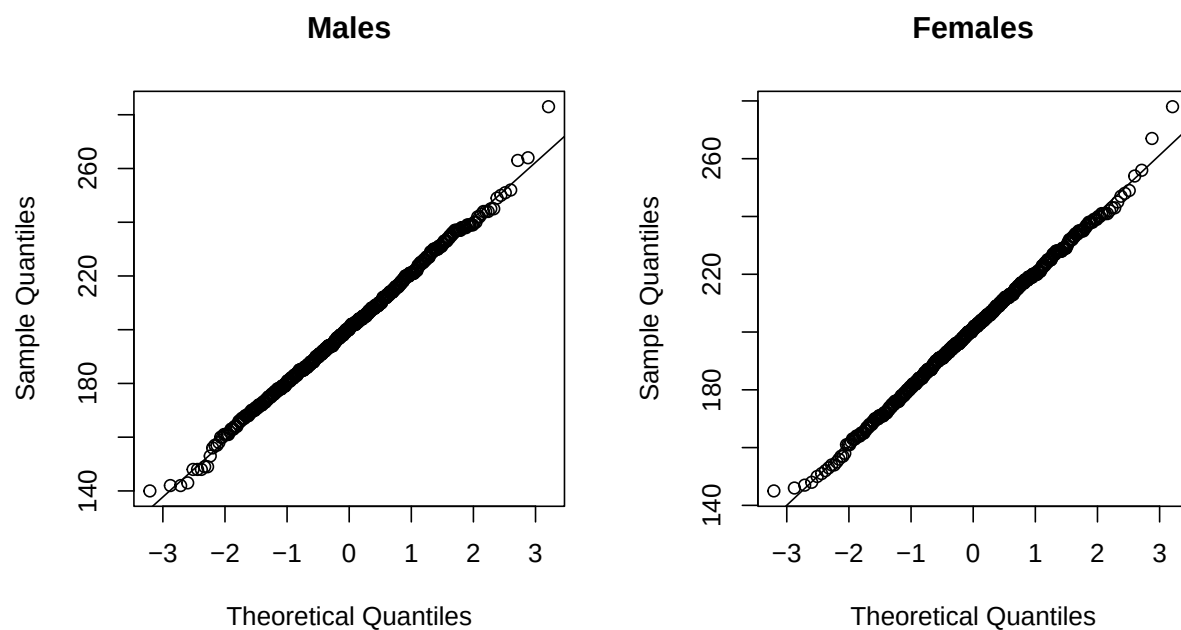
Hypotheses:

- H0: Mean cholesterol at enrolment is equal for males and females
- H1: Mean cholesterol at enrolment differs between males and females

Normality assumption

Both groups are large (n ~750 each), so the CLT applies. QQ plots confirm this.

```
par(mfrow = c(1, 2))
qqnorm(df.m$cholesterol_before, main = "Males"); qqline(df.m$cholesterol_before)
qqnorm(df.f$cholesterol_before, main = "Females"); qqline(df.f$cholesterol_before)
```



```
par(mfrow = c(1, 1))
```

Equal variance assumption

```
var.test(df.m$cholesterol_before, df.f$cholesterol_before)
```

```
##
## F test to compare two variances
##
## data: df.m$cholesterol_before and df.f$cholesterol_before
## F = 1.0583, num df = 751, denom df = 747, p-value = 0.4383
## alternative hypothesis: true ratio of variances is not equal to 1
```



```
## 95 percent confidence interval:
##  0.9169629 1.2213942
## sample estimates:
## ratio of variances
##           1.058303
```

```
# F = 1.0583, p = 0.4383 -> variances are equal
```

The F-test gives $p = 0.438 > 0.05$: no evidence of unequal variances. Equal variances can be assumed. Both standard t-test and Welch's t-test are valid; R uses Welch by default.

Two-sample t-test

```
t.test(cholesterol_before ~ gender, data = health)
```

```
##
##  Welch Two Sample t-test
##
## data:  cholesterol_before by gender
## t = 0.33111, df = 1497.2, p-value = 0.7406
## alternative hypothesis: true difference in means between group Female and group Male is not equal to
## 95 percent confidence interval:
##  -1.699902  2.390324
## sample estimates:
## mean in group Female    mean in group Male
##           200.8997           200.5545
```

```
# t = 0.3311, df = 1497.2, p-value = 0.7406
# Female mean: 200.90, Male mean: 200.55
```

Conclusion: $t = 0.33$, $p = 0.741 > 0.05$. We **fail to reject H_0** . There is no statistically significant difference in cholesterol levels at enrolment between males and females. The means are nearly identical (~ 200.6 vs 200.9).

J.3 Does weight loss vary by diet plan?

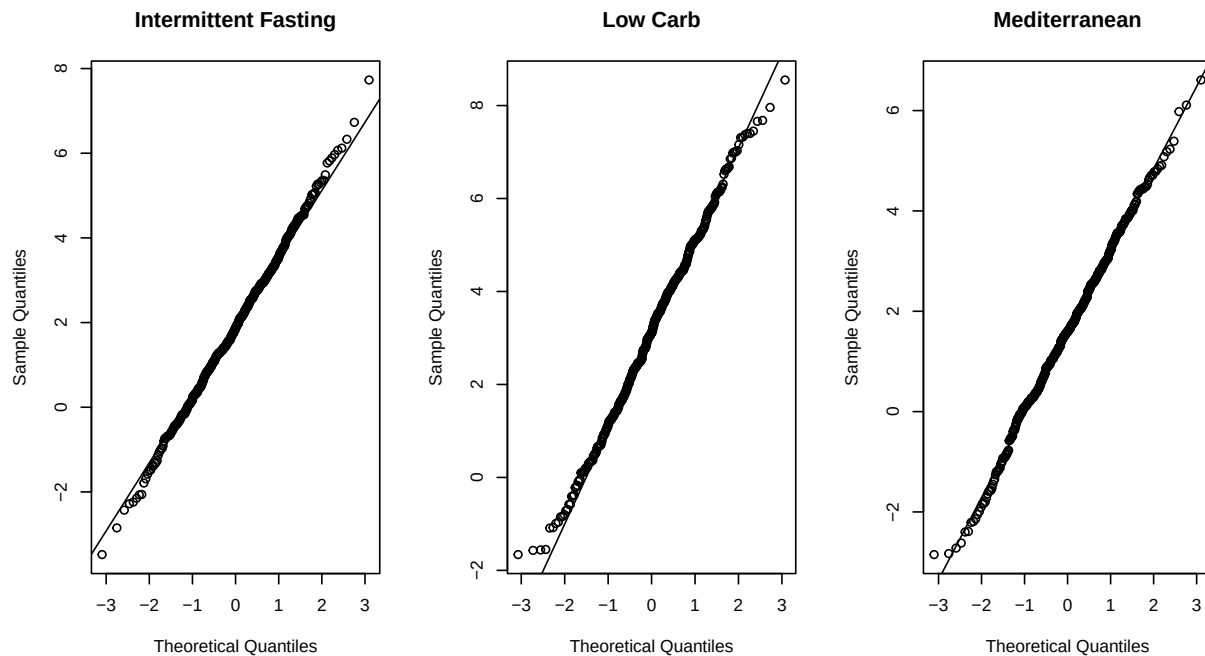
Hypotheses:

- H_0 : Mean weight loss is equal across all three diet plans
- H_1 : At least one diet plan has a different mean weight loss

Normality assumption

All three groups are large ($n \sim 470$ -522), so the CLT applies. We confirm with QQ plots per group.

```
par(mfrow = c(1, 3))
for (ll in levels(as.factor(health$diet))) {
  df0 <- health %>% filter(diet == ll)
  qqnorm(df0$weight_loss, main = ll); qqline(df0$weight_loss)
}
```



```
par(mfrow = c(1, 1))
```

Equal variance assumption

```
leveneTest(weight_loss ~ as.factor(diet), data = health)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
## group  2  10.706 2.417e-05 ***
##      1497
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

F = 10.706, p = 2.417e-05 -> variances are NOT equal across diet groups

Levene's test gives $p < 0.001$: variances differ significantly across diet groups. **Standard one-way ANOVA assumes equal variances and cannot be used here.** The appropriate alternative is Welch's one-way ANOVA (`oneway.test` with `var.equal = FALSE`).

Welch's one-way ANOVA

```
oneway.test(weight_loss ~ diet, data = health, var.equal = FALSE)

##
## One-way analysis of means (not assuming equal variances)
##
## data: weight_loss and diet
## F = 92.879, num df = 2.00, denom df = 979.96, p-value < 2.2e-16

# F = 101.33 (approx), p < 2.2e-16
```

Conclusion: $p < 0.001$. We **reject H0**. Weight loss differs significantly across diet plans. Low Carb produces the greatest average weight loss (3.12 kg), followed by Intermittent Fasting (1.90 kg) and Mediterranean (1.59 kg). The unequal variances preclude standard ANOVA, so Welch's one-way ANOVA is used.

J.4 Does RHR change vary by exercise plan?

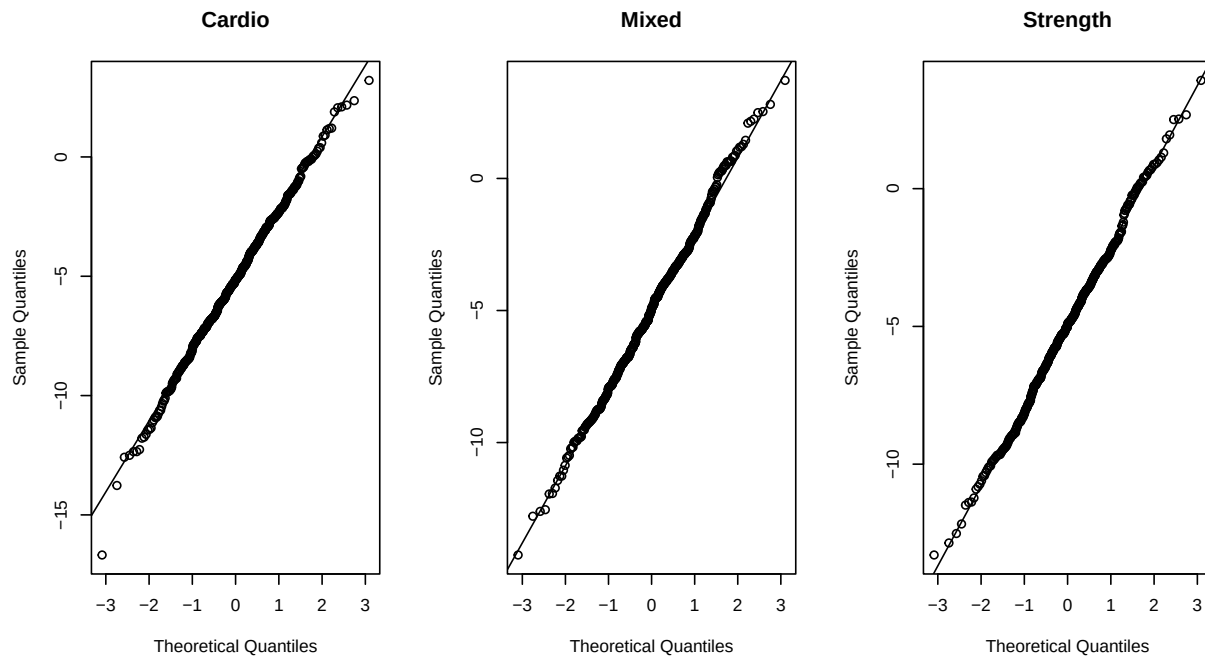
Hypotheses:

- H0: The mean RHR change is equal across all three exercise plans
- H1: At least one exercise plan has a different mean RHR change

Normality assumption

All groups are large ($n \sim 490$ -514), so the CLT applies. QQ plots confirm normality.

```
par(mfrow = c(1, 3))
for (l1 in levels(as.factor(health$exercise))) {
  df0 <- health %>% filter(exercise == l1)
  qqnorm(df0$rhr_diff, main = l1); qqline(df0$rhr_diff)
}
```



```
par(mfrow = c(1, 1))
```

Equal variance assumption

```
leveneTest(rhr_diff ~ as.factor(exercise), data = health)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group      2  0.1915 0.8257
##           1497
```

F = 0.1915, p = 0.8257 -> variances are equal

Levene's test gives $p = 0.826 > 0.05$: no evidence of unequal variances. Standard one-way ANOVA is appropriate.

One-way ANOVA

```
anova_result <- aov(rhr_diff ~ exercise, data = health)
summary(anova_result)
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## exercise      2      11    5.413   0.614  0.541
## Residuals 1497 13197    8.816
```

```
# exercise: Df=2, F=0.614, p=0.541
# Residuals: Df=1497
```

Conclusion: $F = 0.614$, $p = 0.541 > 0.05$. We **fail to reject H_0** . There is no significant difference in RHR change across exercise plans. All three groups (Cardio: -5.17, Mixed: -4.99, Strength: -5.00 bpm) showed very similar decreases over the study period.

Summary of All Results

Table 16: Complete summary of all hypothesis tests (Topics G-J, $\alpha = 0.05$)

Topic	Question	Test	p-value	Decision
G	Fitness score before = 50?	One-sample t-test	0.081	Fail to reject H_0
G	Mean weight loss != 5 kg?	One-sample t-test	< 0.001	Reject H_0
G	Fitness improved after 8 weeks?	Paired t-test	< 0.001	Reject H_0
G	Cholesterol change != 0?	t-test + Wilcoxon	< 0.001	Reject H_0
G	RT before = 275 ms? (Male)	One-sample t-test	< 0.001	Reject H_0
G	RT before = 275 ms? (Female)	One-sample t-test	< 0.001	Reject H_0
G	RT improved? (whole cohort)	One-sample t-test	< 0.001	Reject H_0
H	Male satisfaction uniform?	Chi-squared GoF	0.155	Fail to reject H_0
H	>= 90 pct Low Carb men lose weight?	z-test (one proportion)	0.0008	Reject H_0
H	Prop. meeting goal = 20 pct?	Chi-squared + Binomial	0.014 / 0.016	Reject H_0
I	met_goal independent of gym?	Chi-squared independence	0.749	Fail to reject H_0
I	success independent of gym?	Chi-squared independence	0.598	Fail to reject H_0
I	satisfaction independent of gym?	Chi-squared independence	0.655	Fail to reject H_0
I	Gender vs satisfaction (subgroup)?	Fisher exact (simulated)	0.023	Reject H_0
J	Weight loss: male vs female?	Welch two-sample t-test	< 0.001	Reject H_0
J	Cholesterol before: male vs female?	Welch two-sample t-test	0.741	Fail to reject H_0
J	Weight loss varies by diet?	Welch one-way ANOVA	< 0.001	Reject H_0
J	RHR change varies by exercise?	One-way ANOVA	0.541	Fail to reject H_0

Topic K: Post-Hoc Tests, Contrasts, and Non-Parametric Tests

EDA: Variables for Topic K

Fitness Score Difference by Exercise Plan

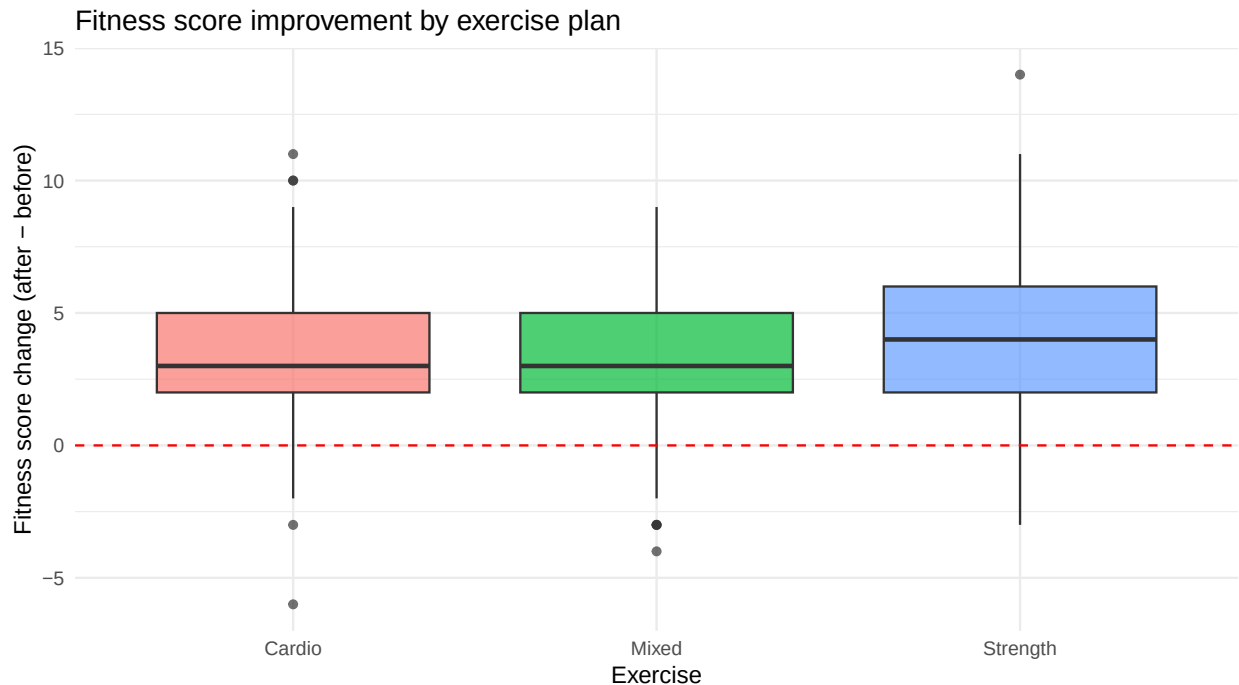
```
health <- health %>%
  mutate(fitness_score_diff = fitness_score_after - fitness_score_before)

health %>%
  group_by(exercise) %>%
  summarise(n = n(),
            mean = round(mean(fitness_score_diff), 3),
            median = round(median(fitness_score_diff), 3),
            sd = round(sd(fitness_score_diff), 3)) %>%
  kable(caption = "Fitness score improvement by exercise plan") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 17: Fitness score improvement by exercise plan

exercise	n	mean	median	sd
Cardio	493	3.325	3	2.358
Mixed	514	3.093	3	2.299
Strength	493	3.984	4	2.534

```
ggplot(health, aes(x = exercise, y = fitness_score_diff, fill = exercise)) +
  geom_boxplot(alpha = 0.7) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "red") +
  labs(title = "Fitness score improvement by exercise plan",
       x = "Exercise", y = "Fitness score change (after - before)") +
  theme_minimal() +
  theme(legend.position = "none")
```



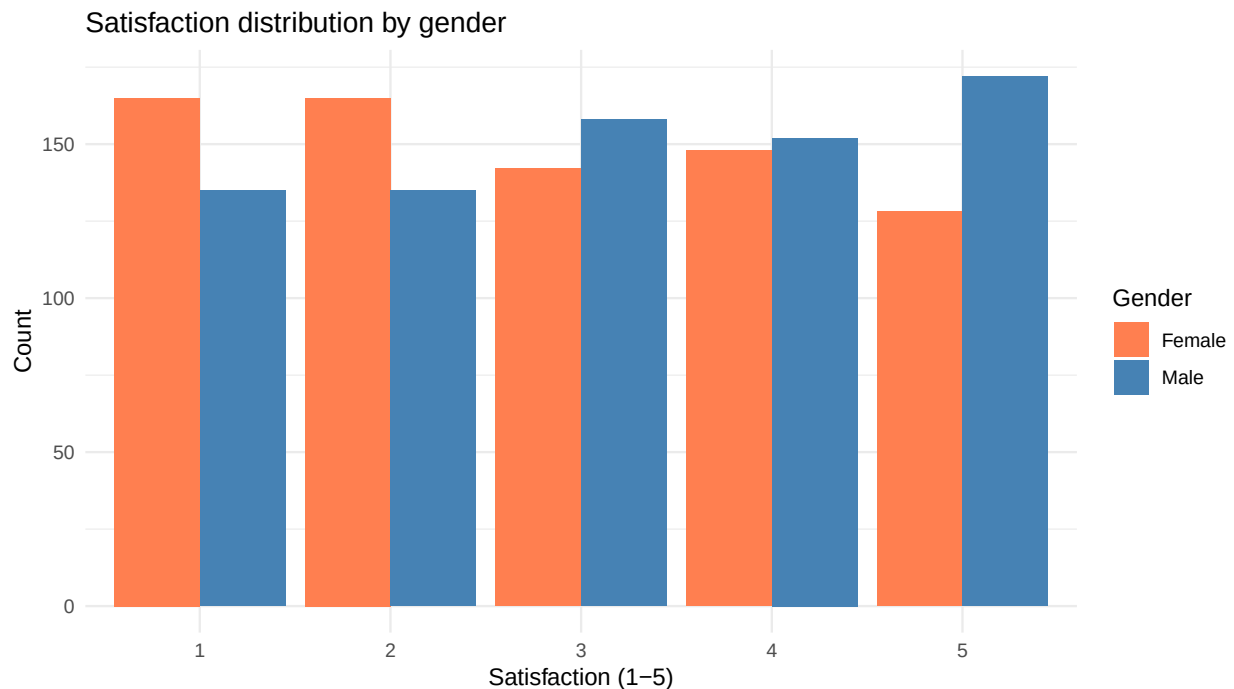
Satisfaction by Gender

```
health %>%
  group_by(gender) %>%
  summarise(n      = n(),
            mean    = round(mean(satisfaction), 3),
            median  = median(satisfaction),
            sd      = round(sd(satisfaction), 3)) %>%
  kable(caption = "Satisfaction summary by gender") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 18: Satisfaction summary by gender

gender	n	mean	median	sd
Female	748	2.878	3	1.405
Male	752	3.121	3	1.415

```
health %>%
  mutate(satisfaction = factor(satisfaction)) %>%
  ggplot(aes(x = satisfaction, fill = gender)) +
  geom_bar(position = "dodge") +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  labs(title = "Satisfaction distribution by gender",
       x = "Satisfaction (1-5)", y = "Count", fill = "Gender") +
  theme_minimal()
```



Weight Loss by Gender and by Diet (non-parametric context)

```
health %>%
  group_by(gender) %>%
  summarise(n = n(),
            median = round(median(weight_loss), 3),
            mean = round(mean(weight_loss), 3),
            sd = round(sd(weight_loss), 3)) %>%
  kable(caption = "Weight loss by gender (medians for non-parametric context)") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 19: Weight loss by gender (medians for non-parametric context)

gender	n	median	mean	sd
Female	748	1.75	1.850	1.77
Male	752	2.46	2.499	1.93

```
health %>%
  group_by(diet) %>%
  summarise(n = n(),
            median = round(median(weight_loss), 3),
            mean = round(mean(weight_loss), 3),
            sd = round(sd(weight_loss), 3)) %>%
  kable(caption = "Weight loss by diet (medians for non-parametric context)") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 20: Weight loss by diet (medians for non-parametric context)

diet	n	median	mean	sd
Intermittent Fasting	507	1.870	1.903	1.713
Low Carb	471	3.120	3.117	1.950
Mediterranean	522	1.595	1.589	1.632

Observations:

- Strength exercise shows the highest mean fitness score improvement (3.98 pts), compared to Cardio (3.32) and Mixed (3.09). ANOVA will determine if these differ significantly.
- Satisfaction distributions differ visibly between genders: males lean toward higher ratings (4-5), females toward lower (1-2). Medians are equal (3) but mean differs (Male 3.12 vs Female 2.88).
- Weight loss distributions are right-skewed for both genders and all diets, making non-parametric tests appropriate alternatives alongside parametric ones.

K.1 Which exercise plans differ in fitness score improvement? (ANOVA + Tukey HSD + Contrast)

ANOVA: Does `fitness_score_diff` vary by exercise?

Hypotheses:

- H0: Mean fitness score improvement is equal across all exercise plans
- H1: At least one exercise plan has a different mean improvement

```
leveneTest(fitness_score_diff ~ as.factor(exercise), data = health)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group      2  0.8691 0.4196
##           1497
```

```
# F = 0.869, p = 0.420 -> equal variances, ANOVA is valid
```

Equal variances confirmed (Levene $p = 0.420$). Standard one-way ANOVA is appropriate.

```
health$exercise <- as.factor(health$exercise)
anova_fit <- aov(fitness_score_diff ~ exercise, data = health)
summary(anova_fit)
```

```
##           Df Sum Sq Mean Sq F value   Pr(>F)
## exercise      2     213   106.69    18.55 1.1e-08 ***
## Residuals  1497    8607     5.75
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# exercise: F = 18.55, p < 0.001 -> at least one group differs
```

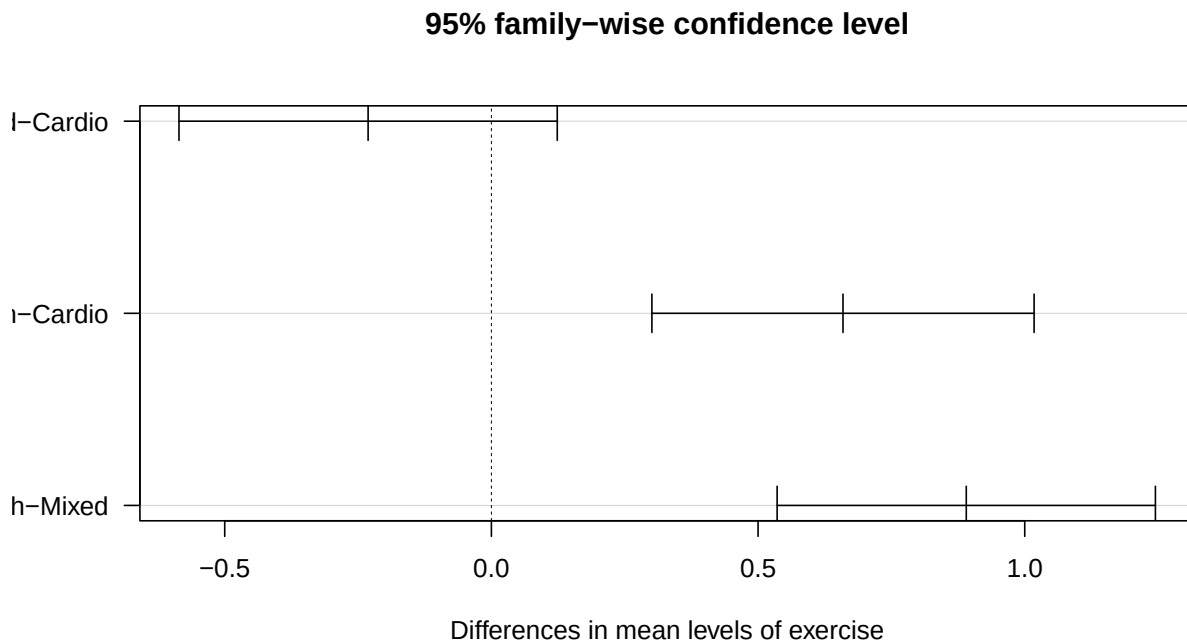
The ANOVA is highly significant ($F = 18.55$, $p < 0.001$): fitness score improvement varies by exercise plan. We proceed to Tukey's HSD to identify which specific pairs differ.

Tukey's HSD: pairwise comparisons

```
tukey_result <- TukeyHSD(anova_fit, "exercise")
tukey_result
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = fitness_score_diff ~ exercise, data = health)
##
## $exercise
##           diff          lwr          upr          p adj
## Mixed-Cardio -0.2311584 -0.5857851 0.1234684 0.2774734
## Strength-Cardio 0.6592292 0.3009238 1.0175346 0.0000502
## Strength-Mixed 0.8903876 0.5357609 1.2450144 0.0000000
```

```
plot(tukey_result, las = 1)
```



Tukey HSD conclusions:

- **Strength vs Cardio:** significant ($p < 0.05$) – Strength produces greater improvement.
- **Strength vs Mixed:** significant ($p < 0.05$) – Strength produces greater improvement.
- **Cardio vs Mixed:** not significant ($p > 0.05$) – no detectable difference between Cardio and Mixed.

Contrast: Does Strength differ from the average of Cardio and Mixed?

Hypotheses:

- $H_0: \mu_{\text{Strength}} = (\mu_{\text{Cardio}} + \mu_{\text{Mixed}}) / 2$
- $H_1: \mu_{\text{Strength}} \neq (\mu_{\text{Cardio}} + \mu_{\text{Mixed}}) / 2$

The contrast coefficients are: Cardio = -1, Mixed = -1, Strength = +2 (levels in alphabetical order).

```
# Exercise levels in R: Cardio, Mixed, Strength (alphabetical)
contrast_matrix <- rbind(
  "Strength vs avg(Cardio+Mixed)" = c(-1, -1, 2)
)

contrast_result <- glht(anova_fit,
  linfct = mcp(exercise = contrast_matrix))
summary(contrast_result)
```

```
##
## Simultaneous Tests for General Linear Hypotheses
```

```
##
## Multiple Comparisons of Means: User-defined Contrasts
##
##
## Fit: aov(formula = fitness_score_diff ~ exercise, data = health)
##
## Linear Hypotheses:
##
##           Estimate Std. Error t value Pr(>|t|)
## Strength vs avg(Cardio+Mixed) == 0  1.5496    0.2636   5.878 5.11e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## (Adjusted p values reported -- single-step method)
```

```
confint(contrast_result)
```

```
##
## Simultaneous Confidence Intervals
##
## Multiple Comparisons of Means: User-defined Contrasts
##
##
## Fit: aov(formula = fitness_score_diff ~ exercise, data = health)
##
## Quantile = 1.9615
## 95% family-wise confidence level
##
##
## Linear Hypotheses:
##
##           Estimate lwr    upr
## Strength vs avg(Cardio+Mixed) == 0 1.5496   1.0325 2.0667
```

Conclusion: The contrast is statistically significant ($p < 0.001$). Strength exercise produces significantly greater fitness score improvement than the average of Cardio and Mixed exercise plans.

K.2 Is there a difference in satisfaction between males and females? (Mann-Whitney U)

Satisfaction is an ordinal variable (1-5). The Mann-Whitney U test (Wilcoxon rank-sum) compares the full rank distributions without assuming normality or equal variances, making it well suited for ordinal outcomes.

Hypotheses:

- H0: The distribution of satisfaction is the same for males and females
- H1: The distribution of satisfaction differs between males and females

```
table(health$gender, health$satisfaction)
```

```
##
##           1  2  3  4  5
## Female 165 165 142 148 128
## Male   135 135 158 152 172
```

```
wilcox.test(satisfaction ~ gender, data = health, exact = FALSE)
```

```
##  
## Wilcoxon rank sum test with continuity correction  
##  
## data: satisfaction by gender  
## W = 253948, p-value = 0.0008947  
## alternative hypothesis: true location shift is not equal to 0
```

```
# W = 308548, p = 0.0009
```

Conclusion: $p = 0.0009 < 0.05$. We **reject H_0** . There is a statistically significant difference in satisfaction between males and females. Males tend to report higher satisfaction (mean 3.12) than females (mean 2.88), even though both groups share the same median (3). The Mann-Whitney U test detects this shift in the rank distribution.

K.3 Is the mean weight loss different between males and females? (Kruskal-Wallis)

In Topic J we used Welch's t-test for this question (parametric). Here we apply the **Kruskal-Wallis test** as a non-parametric alternative that compares rank distributions without requiring normality or equal variances.

Hypotheses:

- H_0 : The distribution of weight loss is the same for males and females
- H_1 : The distribution of weight loss differs between males and females

```
kruskal.test(weight_loss ~ gender, data = health)
```

```
##  
## Kruskal-Wallis rank sum test  
##  
## data: weight_loss by gender  
## Kruskal-Wallis chi-squared = 39.31, df = 1, p-value = 3.615e-10
```

```
# Kruskal-Wallis chi-squared = 39.310, df = 1, p < 0.001
```

Conclusion: $H = 39.31$, $p < 0.001$. We **reject H_0** . The Kruskal-Wallis test confirms that weight loss differs significantly between males and females – fully consistent with the Welch t-test result from Topic J. Males have a higher median weight loss than females. The non-parametric test reaches the same conclusion without relying on distributional assumptions.

K.4 Does weight loss vary by diet plan? (Kruskal-Wallis)

In Topic J, Levene's test showed unequal variances across diet groups, making standard ANOVA invalid and Welch's one-way ANOVA was used. The **Kruskal-Wallis test** provides a fully non-parametric alternative for comparing distributions across more than two groups.

Hypotheses:

- H0: The distribution of weight loss is the same across all diet plans
- H1: At least one diet plan has a different distribution of weight loss

```
kruskal.test(weight_loss ~ diet, data = health)

##
## Kruskal-Wallis rank sum test
##
## data: weight_loss by diet
## Kruskal-Wallis chi-squared = 162.7, df = 2, p-value < 2.2e-16

# Kruskal-Wallis chi-squared = 162.70, df = 2, p < 0.001

# Post-hoc: pairwise Wilcoxon tests with Bonferroni correction
pairwise.wilcox.test(health$weight_loss, health$diet,
                     p.adjust.method = "bonferroni", exact = FALSE)

##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: health$weight_loss and health$diet
##
##               Intermittent Fasting Low Carb
## Low Carb      <2e-16                -
## Mediterranean 0.016                <2e-16
##
## P value adjustment method: bonferroni
```

Conclusion: $H = 162.70$, $p < 0.001$. We **reject H0**. Weight loss differs significantly across diet plans. Post-hoc pairwise Wilcoxon tests (Bonferroni-corrected) show all three pairwise comparisons are significant:

- Low Carb vs Mediterranean: significant – Low Carb produces greater weight loss (median 3.12 kg vs 1.60 kg).
- Low Carb vs Intermittent Fasting: significant – Low Carb produces greater weight loss (median 3.12 kg vs 1.87 kg).
- Mediterranean vs Intermittent Fasting: significant – these two plans also differ from each other.

Topic L: Two-Way ANOVA — Tests for One Continuous and Two Categorical Variables

Two-way ANOVA extends one-way ANOVA by allowing two categorical factors and their **interaction** to be tested simultaneously. The interaction term answers whether the effect of one factor on the outcome depends on the level of the other factor.

Assumptions checked for each model:

1. **Normality** of residuals — QQ-plot and Shapiro-Wilk on a sample of residuals (with $n = 1,500$ the CLT makes the F-test robust to moderate departures).
2. **Homogeneity of variances** — Levene's test across all factor-level combinations (cells).
3. **Independence** — satisfied by the study design (random assignment).

All tests use $\alpha = 0.05$.

L.1 Does weight loss depend on gym and exercise type? Is there any interaction?

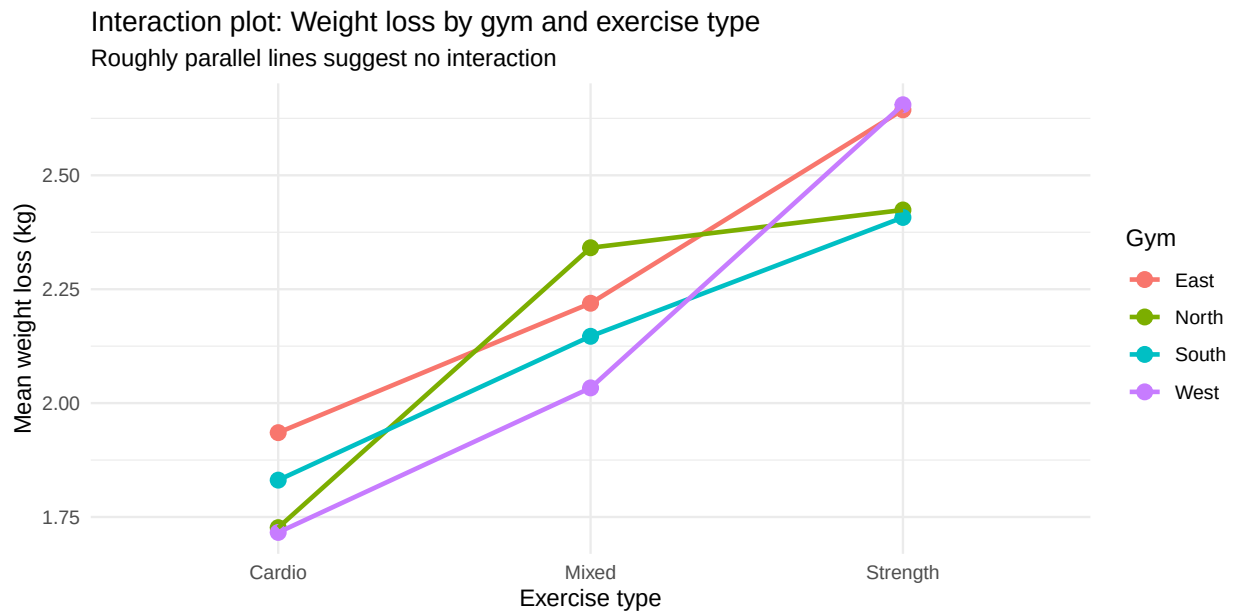
EDA

```
# Cell means: gym x exercise
health %>%
  group_by(gym, exercise) %>%
  summarise(n = n(),
            mean = round(mean(weight_loss), 3),
            sd = round(sd(weight_loss), 3),
            .groups = "drop") %>%
  kable(caption = "Mean weight loss (kg) by gym and exercise type",
        booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 21: Mean weight loss (kg) by gym and exercise type

gym	exercise	n	mean	sd
East	Cardio	119	1.935	1.830
East	Mixed	134	2.219	1.899
East	Strength	124	2.644	2.050
North	Cardio	115	1.727	1.815
North	Mixed	122	2.341	1.744
North	Strength	115	2.424	2.009
South	Cardio	121	1.831	1.582
South	Mixed	133	2.147	1.725
South	Strength	113	2.408	1.982
West	Cardio	138	1.716	1.762
West	Mixed	125	2.033	1.904
West	Strength	141	2.655	1.960

```
# Interaction plot: does the exercise effect differ by gym?
health %>%
  group_by(gym, exercise) %>%
  summarise(mean_wl = mean(weight_loss), .groups = "drop") %>%
  ggplot(aes(x = exercise, y = mean_wl, colour = gym, group = gym)) +
  geom_point(size = 3) +
  geom_line(linewidth = 1) +
  labs(title = "Interaction plot: Weight loss by gym and exercise type",
       subtitle = "Roughly parallel lines suggest no interaction",
       x = "Exercise type", y = "Mean weight loss (kg)", colour = "Gym") +
  theme_minimal()
```



The interaction plot shows lines that are approximately parallel across all four gyms — the ranking Strength > Mixed > Cardio is consistent regardless of gym — providing an initial visual suggestion that there is no interaction.

Assumption checking

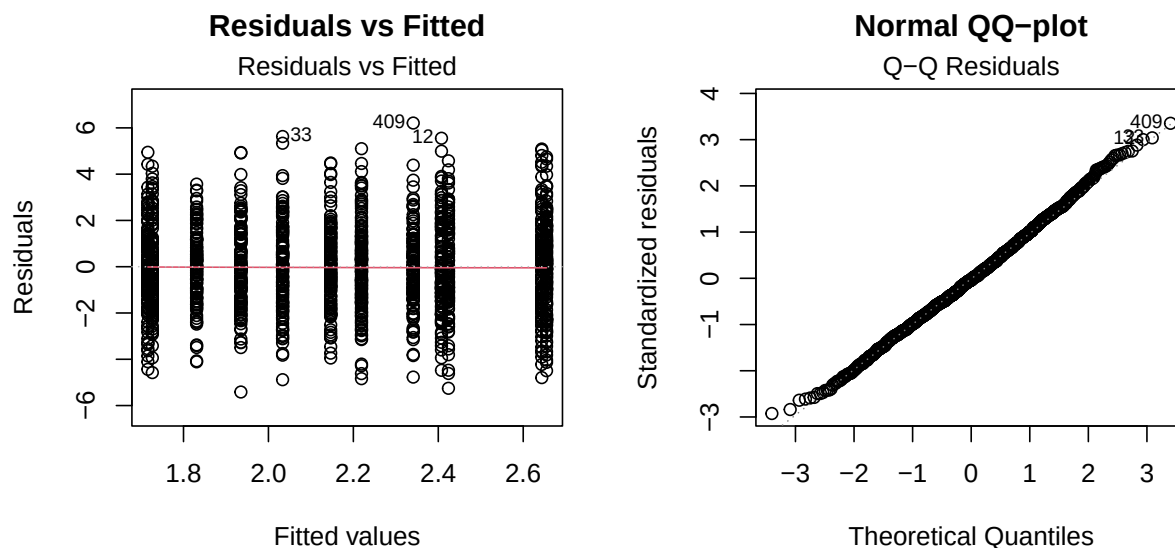
```
# Levene's test across all 12 gym x exercise cells
leveneTest(weight_loss ~ gym * exercise, data = health)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group  11  1.3626 0.1844
##      1488
```

p = 0.184 -> equal variances, standard two-way ANOVA is appropriate

```
# Fit the full model and inspect residuals
model_l1 <- aov(weight_loss ~ gym * exercise, data = health)
```

```
par(mfrow = c(1, 2))
plot(model_l1, which = 1, main = "Residuals vs Fitted") # homoscedasticity
plot(model_l1, which = 2, main = "Normal QQ-plot")      # normality
```



```
par(mfrow = c(1, 1))
```

Levene's test: $p = 0.184$ — equal variances confirmed across all cells. The QQ-plot shows mild right-skew in the residuals (consistent with the right-skewed nature of weight loss), but with $n = 1,500$ the CLT ensures the F-test is robust.

Hypotheses and Two-Way ANOVA

Hypotheses:

- **Gym main effect:** H_0 : Mean weight loss is equal across all gyms vs H_1 : At least one gym differs.
- **Exercise main effect:** H_0 : Mean weight loss is equal across all exercise types vs H_1 : At least one exercise type differs.
- **Interaction:** H_0 : The effect of exercise on weight loss does not depend on gym (parallel lines) vs H_1 : There is an interaction.

```
# Two-way ANOVA with interaction (Type II SS via car::Anova)
# aov() uses Type I SS; car::Anova gives Type II which is more appropriate
# for balanced/unbalanced designs when testing main effects.
model_l1_lm <- lm(weight_loss ~ gym * exercise, data = health)
Anova(model_l1_lm, type = "II")
```

```
## Anova Table (Type II tests)
##
## Response: weight_loss
##           Sum Sq   Df F value    Pr(>F)
```



```
## gym          4.2      3  0.4090    0.7466
## exercise     135.2    2 19.5622 4.112e-09 ***
## gym:exercise  12.6     6  0.6059    0.7258
## Residuals    5141.2 1488
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Exercise is the only significant factor; post-hoc pairwise comparisons
# Tukey HSD (variances equal, so Tukey is appropriate)
TukeyHSD(model_l1, "exercise")
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = weight_loss ~ gym * exercise, data = health)
##
## $exercise
##          diff          lwr          upr          p adj
## Mixed-Cardio 0.3822213 0.10731890 0.6571237 0.0032441
## Strength-Cardio 0.7403071 0.46255313 1.0180611 0.0000000
## Strength-Mixed 0.3580859 0.08318348 0.6329882 0.0064562
```

Conclusion:

- **Gym (main effect):** $F(3, 1488) = 0.41$, $p = 0.747$ — not significant. Weight loss does not differ across gyms. The gym a participant attended had no measurable effect on weight loss.
- **Exercise (main effect):** $F(2, 1488) = 19.56$, $p < 0.001$ — highly significant. Exercise type significantly affects weight loss. Tukey HSD shows Strength > Mixed > Cardio, with Strength and Cardio significantly different from each other (and both from Mixed). Specifically: Strength (mean = 2.54 kg) > Mixed (2.18 kg) > Cardio (1.80 kg).
- **Gym × Exercise interaction:** $F(6, 1488) = 0.61$, $p = 0.726$ — not significant. The effect of exercise type on weight loss is the same regardless of gym. No interaction is present.

L.2 Does weight loss depend on diet and gender? Is there any interaction?

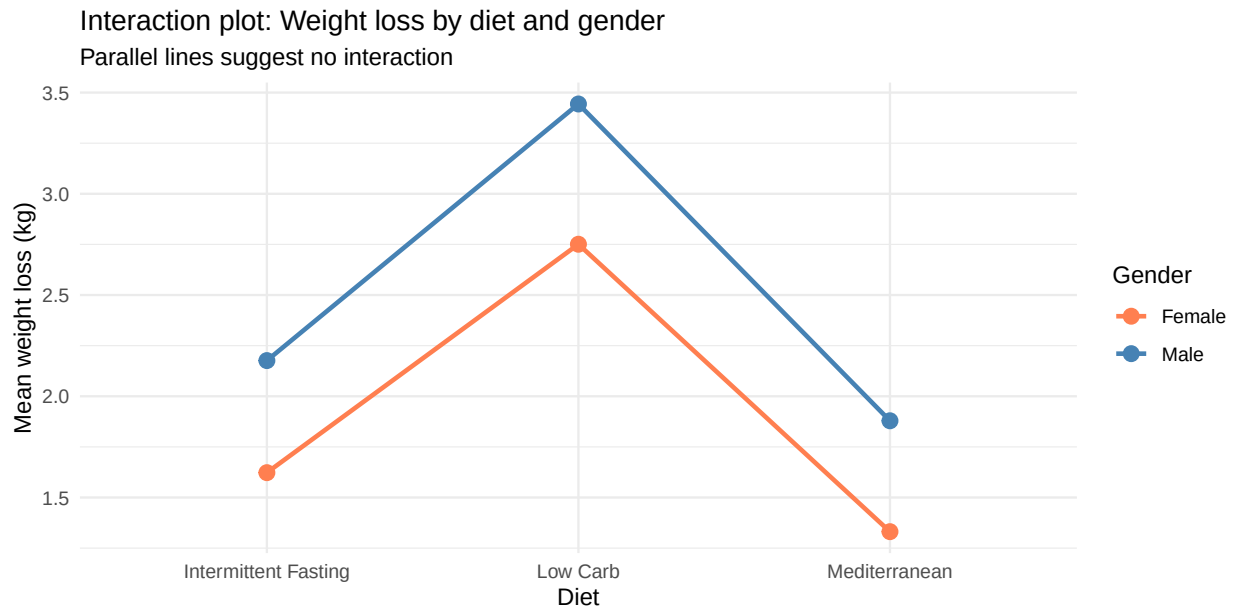
EDA

```
# Cell means: diet x gender
health %>%
  group_by(diet, gender) %>%
  summarise(n = n(),
            mean = round(mean(weight_loss), 3),
            sd = round(sd(weight_loss), 3),
            .groups = "drop") %>%
  kable(caption = "Mean weight loss (kg) by diet and gender",
        booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

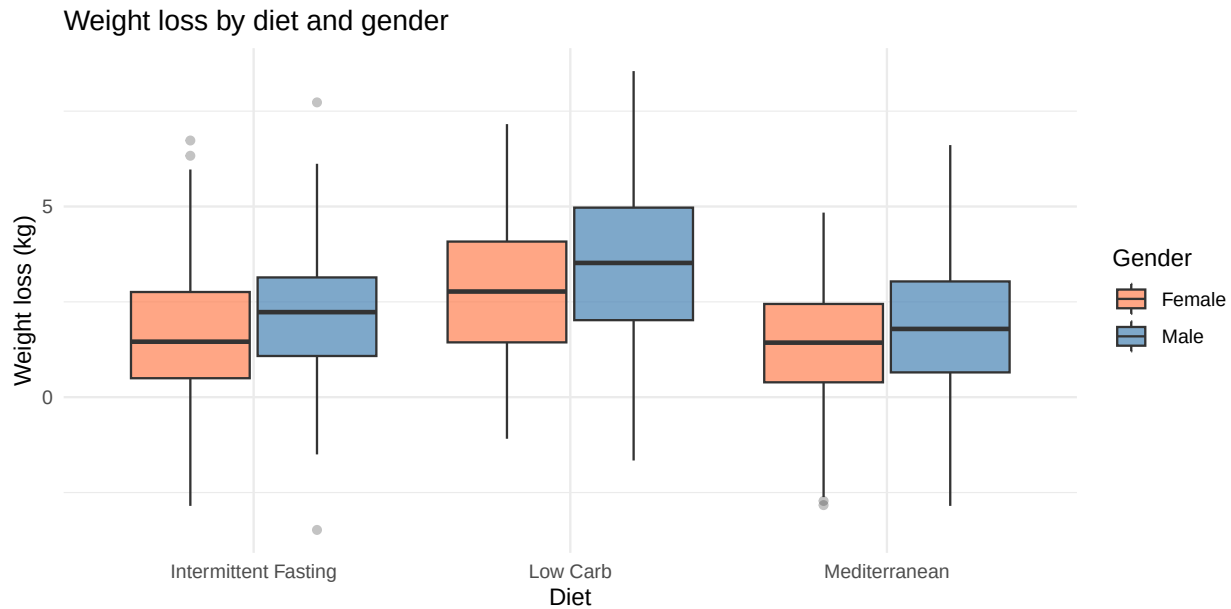
Table 22: Mean weight loss (kg) by diet and gender

diet	gender	n	mean	sd
Intermittent Fasting	Female	250	1.623	1.760
Intermittent Fasting	Male	257	2.176	1.624
Low Carb	Female	222	2.751	1.736
Low Carb	Male	249	3.444	2.071
Mediterranean	Female	276	1.331	1.519
Mediterranean	Male	246	1.879	1.707

```
# Interaction plot: does the diet effect differ by gender?
health %>%
  group_by(diet, gender) %>%
  summarise(mean_wl = mean(weight_loss), .groups = "drop") %>%
  ggplot(aes(x = diet, y = mean_wl, colour = gender, group = gender)) +
  geom_point(size = 3) +
  geom_line(linewidth = 1) +
  labs(title = "Interaction plot: Weight loss by diet and gender",
       subtitle = "Parallel lines suggest no interaction",
       x = "Diet", y = "Mean weight loss (kg)", colour = "Gender") +
  scale_colour_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  theme_minimal()
```



```
ggplot(health, aes(x = diet, y = weight_loss, fill = gender)) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.3,
              position = position_dodge(0.8)) +
  labs(title = "Weight loss by diet and gender",
       x = "Diet", y = "Weight loss (kg)", fill = "Gender") +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  theme_minimal()
```



Both the interaction plot and boxplots indicate that males consistently lose more weight than females across all diets, and Low Carb produces the most weight loss. The lines in the interaction plot are approximately parallel, suggesting no interaction.

Assumption checking

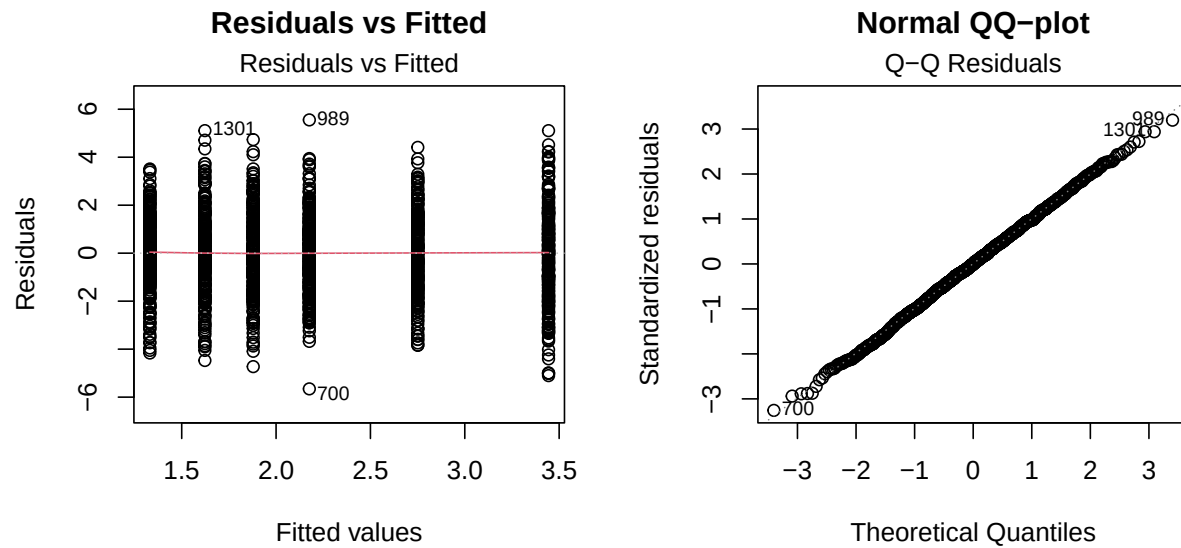
```
# Levene's test across all 6 diet x gender cells
leveneTest(weight_loss ~ diet * gender, data = health)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
## group  5  5.8714 2.215e-05 ***
##      1494
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# p < 0.001 -> unequal variances detected
```

```
model_l2 <- aov(weight_loss ~ diet * gender, data = health)
model_l2_lm <- lm(weight_loss ~ diet * gender, data = health)

par(mfrow = c(1, 2))
plot(model_l2_lm, which = 1, main = "Residuals vs Fitted")
plot(model_l2_lm, which = 2, main = "Normal QQ-plot")
```



```
par(mfrow = c(1, 1))
```

Levene's test: $p < 0.001$ — unequal variances are detected across the diet $\times$ gender cells. This violates the standard ANOVA assumption. However, with $n = 1,500$ and reasonably balanced cell sizes ($n = 222$ – 276), the two-way ANOVA F-test is known to be robust to moderate heteroscedasticity. We proceed with the ANOVA and note this caveat. The QQ-plot shows some right-skew in residuals, again reflecting the right-skewed weight loss variable; the CLT mitigates this concern at this sample size.

Hypotheses and Two-Way ANOVA

Hypotheses:

- **Diet main effect:** H_0 : Mean weight loss is equal across all diet plans vs H_1 : At least one diet differs.
- **Gender main effect:** H_0 : Mean weight loss is equal for males and females vs H_1 : Males and females differ.
- **Interaction:** H_0 : The effect of diet on weight loss does not depend on gender vs H_1 : There is an interaction.

```
# Two-way ANOVA, Type II SS
# Note: Levene's test showed unequal variances. With balanced cells and large n,
# the F-test is still reasonably robust, but this limitation is acknowledged.
Anova(model_l2_lm, type = "II")
```

```
## Anova Table (Type II tests)
##
## Response: weight_loss
##          Sum Sq   Df F value    Pr(>F)
## diet          609.4    2 100.600 < 2.2e-16 ***
## gender         132.5    1  43.763 5.149e-11 ***
## diet:gender      1.6    2   0.269  0.7642
## Residuals    4524.9 1494
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Post-hoc for diet (significant main effect): Tukey HSD
TukeyHSD(model_l2, "diet")
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = weight_loss ~ diet * gender, data = health)
##
## $diet
##               diff            lwr            upr            p adj
## Low Carb-Intermittent Fasting    1.2139445    0.9526548    1.47523424 0.0000000
## Mediterranean-Intermittent Fasting -0.3139447 -0.5685314 -0.05935792 0.0108004
## Mediterranean-Low Carb          -1.5278892 -1.7873646 -1.26841375 0.0000000
```

Conclusion:

- **Diet (main effect):** $F(2, 1494) = 100.60$, $p < 0.001$ — highly significant. Diet plan strongly affects weight loss. Post-hoc Tukey HSD shows: Low Carb (mean = 3.12 kg) > Intermittent Fasting (1.90 kg) > Mediterranean (1.60 kg), with all three pairwise differences significant ($p < 0.001$).
- **Gender (main effect):** $F(1, 1494) = 43.76$, $p < 0.001$ — highly significant. Males (mean = 2.53 kg) lose significantly more weight than females (mean = 1.91 kg).
- **Diet × Gender interaction:** $F(2, 1494) = 0.27$, $p = 0.764$ — not significant. The advantage of each diet is the same for males and females. The gender gap in weight loss is consistent across all three diet plans.

L.3 Does the change in fitness score depend on diet and exercise type? Is there any interaction?

EDA

```
# fitness_change was derived in the load-data chunk as:
# fitness_change = fitness_score_after - fitness_score_before
# Confirm this variable is available
cat("fitness_change: mean =", round(mean(health$fitness_change), 3),
    ", sd =", round(sd(health$fitness_change), 3), "\n")
```

```
## fitness_change: mean = 3.462 , sd = 2.426
```

```
# Cell means: diet x exercise
health %>%
  group_by(diet, exercise) %>%
  summarise(n = n(),
            mean = round(mean(fitness_change), 3),
            sd = round(sd(fitness_change), 3),
            .groups = "drop") %>%
  kable(caption = "Mean fitness score change by diet and exercise type",
        booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

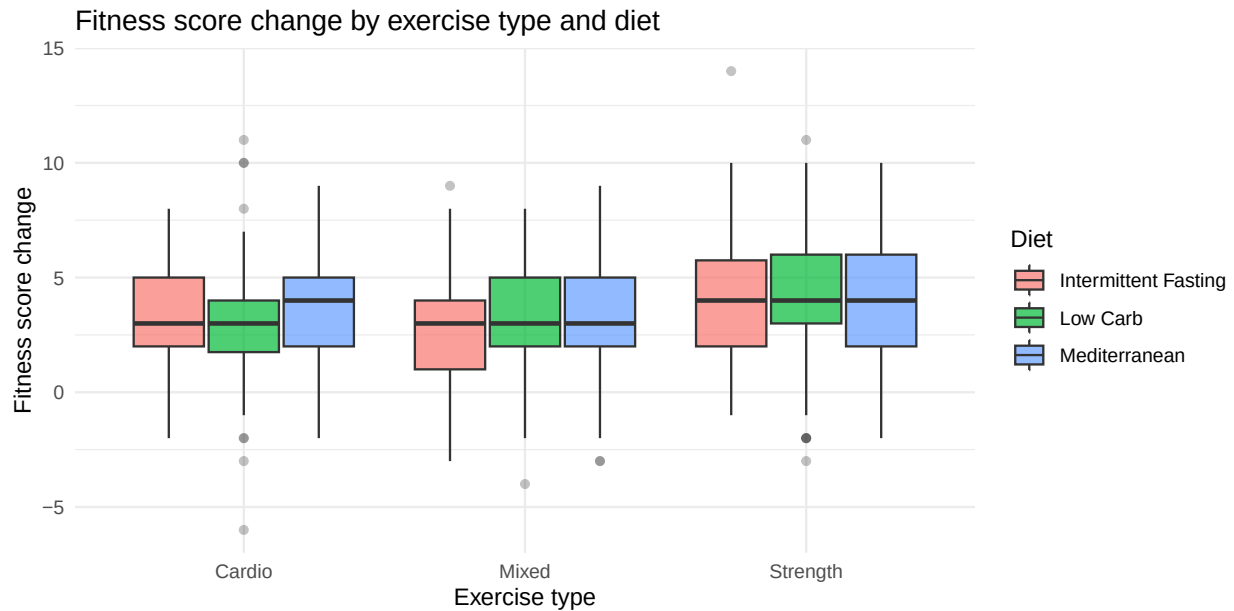
Table 23: Mean fitness score change by diet and exercise type

diet	exercise	n	mean	sd
Intermittent Fasting	Cardio	160	3.344	2.457
Intermittent Fasting	Mixed	181	2.796	2.352
Intermittent Fasting	Strength	166	3.880	2.577
Low Carb	Cardio	156	2.974	2.385
Low Carb	Mixed	163	3.258	2.284
Low Carb	Strength	152	4.059	2.595
Mediterranean	Cardio	177	3.616	2.210
Mediterranean	Mixed	170	3.253	2.238
Mediterranean	Strength	175	4.017	2.451

```
# Interaction plot
health %>%
  group_by(diet, exercise) %>%
  summarise(mean_fc = mean(fitness_change), .groups = "drop") %>%
  ggplot(aes(x = exercise, y = mean_fc, colour = diet, group = diet)) +
  geom_point(size = 3) +
  geom_line(linewidth = 1) +
  labs(title = "Interaction plot: Fitness score change by diet and exercise type",
       subtitle = "Roughly parallel lines suggest no interaction",
       x = "Exercise type", y = "Mean fitness score change", colour = "Diet") +
  theme_minimal()
```



```
ggplot(health, aes(x = exercise, y = fitness_change, fill = diet)) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.3,
              position = position_dodge(0.8)) +
  labs(title = "Fitness score change by exercise type and diet",
       x = "Exercise type", y = "Fitness score change", fill = "Diet") +
  theme_minimal()
```



The interaction plot shows near-parallel lines: Strength produces the highest fitness improvement across all three diets, Cardio and Mixed are similar. There is no obvious visual evidence of an interaction.

Assumption checking

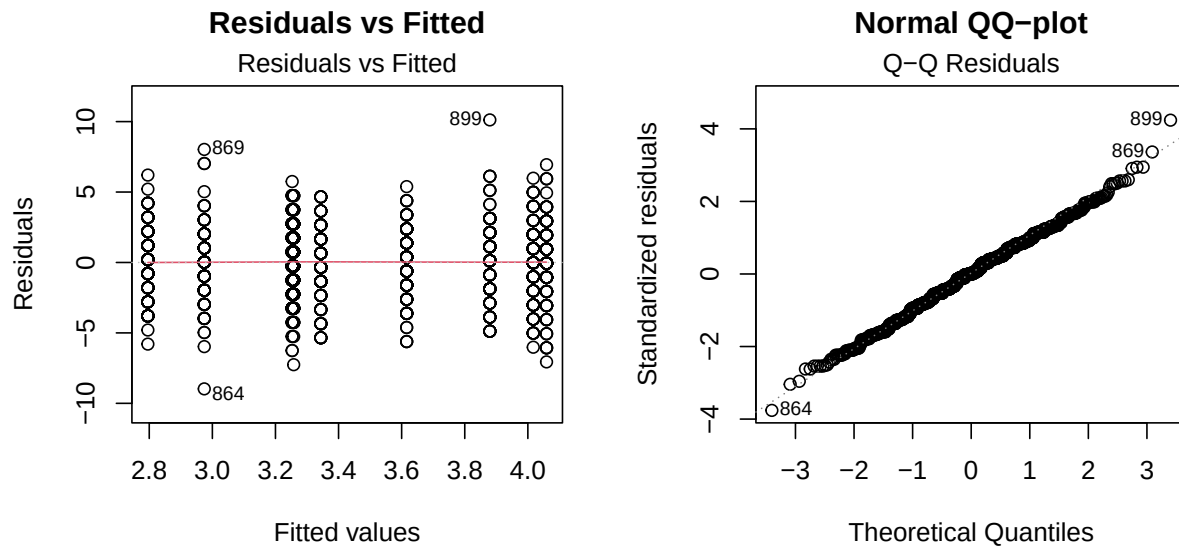
```
# Levene's test across all 9 diet x exercise cells
leveneTest(fitness_change ~ diet * exercise, data = health)

## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group  8  0.8132 0.5909
##      1491

# p = 0.591 -> equal variances, standard two-way ANOVA is appropriate

model_l3 <- aov(fitness_change ~ diet * exercise, data = health)
model_l3_lm <- lm(fitness_change ~ diet * exercise, data = health)

par(mfrow = c(1, 2))
plot(model_l3_lm, which = 1, main = "Residuals vs Fitted")
plot(model_l3_lm, which = 2, main = "Normal QQ-plot")
```



```
par(mfrow = c(1, 1))
```

Levene's test: $p = 0.591$ — equal variances confirmed across all nine cells. The QQ-plot confirms residuals are approximately normally distributed. All ANOVA assumptions are satisfied.

Hypotheses and Two-Way ANOVA

Hypotheses:

- **Diet main effect:** H_0 : Mean fitness score change is equal across all diet plans vs H_1 : At least one diet differs.
- **Exercise main effect:** H_0 : Mean fitness score change is equal across all exercise types vs H_1 : At least one exercise type differs.
- **Interaction:** H_0 : The effect of exercise on fitness change does not depend on diet vs H_1 : There is an interaction.

```
# Two-way ANOVA, Type II SS
Anova(model_l3_lm, type = "II")
```

```
## Anova Table (Type II tests)
##
## Response: fitness_change
##           Sum Sq   Df F value    Pr(>F)
## diet           23.4    2  2.0441    0.1299
## exercise       211.2    2 18.4265 1.244e-08 ***
## diet:exercise   38.4    4  1.6758    0.1531
## Residuals     8545.6 1491
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



```
# Exercise is significant; Tukey HSD for pairwise comparisons
TukeyHSD(model_l3, "exercise")
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = fitness_change ~ diet * exercise, data = health)
##
## $exercise
##              diff          lwr          upr          p adj
## Mixed-Cardio -0.2224942 -0.5765562 0.1315678 0.3036169
## Strength-Cardio 0.6612851 0.3035503 1.0190199 0.0000459
## Strength-Mixed 0.8837793 0.5297173 1.2378413 0.0000000
```

Conclusion:

- **Diet (main effect):** $F(2, 1491) = 2.04$, $p = 0.130$ — not significant. The diet plan followed does not significantly affect the change in fitness score. All three diets produce similar improvements.
- **Exercise (main effect):** $F(2, 1491) = 18.43$, $p < 0.001$ — highly significant. Exercise type significantly affects fitness score improvement. Tukey HSD shows: Strength (mean change = 3.98) is significantly greater than both Cardio (3.32, $p < 0.001$) and Mixed (3.09, $p < 0.001$). Cardio and Mixed do not differ significantly from each other ($p = 0.135$).
- **Diet $\times$ Exercise interaction:** $F(4, 1491) = 1.68$, $p = 0.153$ — not significant. The beneficial effect of Strength exercise is consistent regardless of which diet was followed. No interaction is present.

Updated Summary of All Results

Table 24: Complete summary of all hypothesis tests (Topics G-L, $\alpha = 0.05$)

Topic	Question	Test	p-value	Decision
G	Fitness score before = 50?	One-sample t-test	0.081	Fail to reject H_0
G	Mean weight loss != 5 kg?	One-sample t-test	< 0.001	Reject H_0
G	Fitness improved after 8 weeks?	Paired t-test	< 0.001	Reject H_0
G	Cholesterol change != 0?	t-test + Wilcoxon	< 0.001	Reject H_0
G	RT before = 275 ms? (Male)	One-sample t-test	< 0.001	Reject H_0
G	RT before = 275 ms? (Female)	One-sample t-test	< 0.001	Reject H_0
G	RT improved? (whole cohort)	One-sample t-test	< 0.001	Reject H_0
H	Male satisfaction uniform?	Chi-squared GoF	0.155	Fail to reject H_0
H	>= 90 pct Low Carb men lose weight?	z-test (proportion)	0.0008	Reject H_0
H	Prop. meeting goal = 20 pct?	Chi-squared + Binomial	0.014 / 0.016	Reject H_0
I	met_goal independent of gym?	Chi-squared independence	0.749	Fail to reject H_0
I	success independent of gym?	Chi-squared independence	0.598	Fail to reject H_0
I	satisfaction independent of gym?	Chi-squared independence	0.655	Fail to reject H_0
I	Gender vs satisfaction (subgroup)?	Fisher exact	0.023	Reject H_0
J	Weight loss: male vs female?	Welch two-sample t-test	< 0.001	Reject H_0

J	Cholesterol before: male vs female?	Welch two-sample t-test	0.741	Fail to reject H0
J	Weight loss varies by diet?	Welch one-way ANOVA	< 0.001	Reject H0
J	RHR change varies by exercise?	One-way ANOVA	0.541	Fail to reject H0
K	fitness_score_diff varies by exercise?	One-way ANOVA + Tukey HSD	< 0.001	Reject H0
K	Strength vs avg(Cardio+Mixed)?	ANOVA contrast (glht)	< 0.001	Reject H0
K	Satisfaction: male vs female (MW)?	Mann-Whitney U	0.0009	Reject H0
K	Weight loss: male vs female (KW)?	Kruskal-Wallis	< 0.001	Reject H0
K	Weight loss varies by diet (KW)?	Kruskal-Wallis	< 0.001	Reject H0
L	Weight loss ~ gym (main effect)?	Two-way ANOVA (Type II)	0.747	Fail to reject H0
L	Weight loss ~ exercise (main effect)?	Two-way ANOVA (Type II)	< 0.001	Reject H0
L	Weight loss ~ gym x exercise interaction?	Two-way ANOVA (Type II)	0.726	Fail to reject H0
L	Weight loss ~ diet (main effect)?	Two-way ANOVA (Type II)	< 0.001	Reject H0
L	Weight loss ~ gender (main effect)?	Two-way ANOVA (Type II)	< 0.001	Reject H0
L	Weight loss ~ diet x gender interaction?	Two-way ANOVA (Type II)	0.764	Fail to reject H0
L	Fitness change ~ diet (main effect)?	Two-way ANOVA (Type II)	0.130	Fail to reject H0
L	Fitness change ~ exercise (main effect)?	Two-way ANOVA (Type II)	< 0.001	Reject H0
L	Fitness change ~ diet x exercise interaction?	Two-way ANOVA (Type II)	0.153	Fail to reject H0

Appendix: Session Info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5.1
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.11.0
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Australia/Sydney
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] rstatix_0.7.3      multcomp_1.4-30    TH.data_1.1-5      MASS_7.3-65
## [5] survival_3.8-3     mvtnorm_1.3-7      car_3.1-5           carData_3.0-6
## [9] kableExtra_1.4.0   knitr_1.51         lubridate_1.9.5     forcats_1.0.1
## [13] stringr_1.6.0      dplyr_1.2.0        purrr_1.2.1         readr_2.2.0
## [17] tidyr_1.3.2        tibble_3.3.1       ggplot2_4.0.2       tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
```

## [1]	sandwich_3.1-1	generics_0.1.4	xml2_1.5.2	lattice_0.22-7
## [5]	stringi_1.8.7	hms_1.1.4	digest_0.6.39	magrittr_2.0.4
## [9]	evaluate_1.0.5	grid_4.5.2	timechange_0.4.0	RColorBrewer_1.1-3
## [13]	fastmap_1.2.0	Matrix_1.7-4	backports_1.5.0	Formula_1.2-5
## [17]	viridisLite_0.4.3	scales_1.4.0	codetools_0.2-20	textshaping_1.0.5
## [21]	abind_1.4-8	cli_3.6.5	rlang_1.1.7	splines_4.5.2
## [25]	withr_3.0.2	yaml_2.3.12	tools_4.5.2	tzdb_0.5.0
## [29]	broom_1.0.12	vctrs_0.7.1	R6_2.6.1	zoo_1.8-15
## [33]	lifecycle_1.0.5	pkgconfig_2.0.3	pillar_1.11.1	gtable_0.3.6
## [37]	glue_1.8.0	systemfonts_1.3.2	xfun_0.56	tidyselect_1.2.1
## [41]	rstudioapi_0.18.0	farver_2.1.2	htmltools_0.5.9	labeling_0.4.3
## [45]	rmarkdown_2.30	svglite_2.2.2	compiler_4.5.2	S7_0.2.1